

A variational-damage-based thermo-mechanical phase-field model for fracture analysis of electronic packaging structures

Yanpeng Gong ^{a, }, Letong Zhang ^a, Ya Duan ^{b, }, Tong An ^a, Xiaoying Zhuang ^{c, }, Timon Rabczuk ^{a, b, *}

^a Department of Mechanics, Beijing University of Technology, Beijing, 100124, China

^b Institute of Structural Mechanics, Bauhaus University Weimar, 99423 Weimar, Germany

^c Chair of Computational Science and Simulation Technology, Institute of Photonics, Department of Mathematics and Physics, Leibniz University Hannover, 30167 Hannover, Germany

^d Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai, 200092, China

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ABSTRACT

The transformed phase-field cohesive zone model (Tpfczm) exploits a bridge relation between the unified phase-field theory and the variational damage model, enabling the constitutive parameters to be derived without the approximation required in the PF-CZM, thereby enhancing convergence robustness while retaining length-scale insensitivity. However, the Tpfczm has so far been limited to purely mechanical fracture problems. In this work, the Tpfczm is extended to thermo-mechanically coupled fracture. The proposed framework incorporates thermal strains into the elastic constitutive relation, introduces a damage-dependent thermal conductivity to capture the thermal barrier effect of cracks, and adopts a tension-shear energy decomposition to account for shear-dominated failure modes. The resulting three-field coupled model — governing the displacement, temperature, and damage fields simultaneously — inherits the improved convergence robustness and length-scale insensitivity of the original Tpfczm. A benchmark comparison against the PF-CZM further confirms the efficiency advantage of the Tpfczm in terms of iteration count and computational time. The proposed framework is applied to a series of representative electronic packaging problems involving intermetallic compound (IMC) layer fracture and through-glass via (TGV) thermo-mechanical damage. The numerical examples illustrate coefficient of thermal expansion (CTE)-mismatch-driven crack-evolution patterns in representative electronic packaging structures, including interfacial cracking, void-induced changes in crack paths, and progressive interfacial fracture.

1. Introduction

The reliability of electronic packaging interconnections has become a critical concern as semiconductor devices continue to advance toward higher integration density [1,2]. In modern packaging architectures, particularly 2.5D/3D integration using through-silicon vias (TSVs) and through-glass vias (TGVs), multiple material layers with significantly different coefficients of thermal expansion (CTE) are bonded together to form the interconnection structure. During manufacturing processes such as reflow soldering and annealing, as well as during in-service thermal cycling and power cycling, the CTE mismatch among constituent materials — including solder alloys, intermetallic compound (IMC) layers, copper pads, silicon dies, and glass substrates — generates substantial

* Corresponding author.

E-mail address: timon.rabczuk@uni-weimar.de (T. Rabczuk).

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Nomenclature

Symbol	Description
<i>Field and thermo-mechanical quantities</i>	
d	Phase-field damage variable
$\mathbf{u}$	Displacement vector
T	Temperature
$\boldsymbol{\sigma}$	Cauchy stress tensor
$\boldsymbol{\sigma}_0$	Effective stress tensor of the undamaged material
$\boldsymbol{\epsilon}$	Total strain tensor
$\boldsymbol{\epsilon}^e$	Elastic strain tensor
$\boldsymbol{\epsilon}^{th}$	Thermal strain tensor
T_0	Stress-free reference temperature
Φ_0	Elastic strain energy density of the undamaged material
Φ_v	Stored energy density of the damaged material
$\mathcal{Y}_v$	Damage-driving force
$\mathbf{g}$	Body-force density
$\bar{\mathbf{i}}$	Prescribed traction
$\mathbf{q}$	Heat-flux vector
$\bar{q}$	Prescribed heat flux on the Neumann boundary
Q_h	Volumetric heat source
$\Omega, \Gamma, \mathcal{B}$	Body domain, sharp crack surface, and localization band, respectively
<i>Phase-field and fracture quantities</i>	
$\gamma(d)$	Crack surface density function
$\alpha_v(d)$	Crack geometric function
$\omega_v(d)$	Energy degradation function
$\varphi_v(d), \varphi_p(d)$	Rational degradation functions of the variational damage and unified phase-field models, respectively
$Q(d), P(d)$	Polynomial functions used in the rational degradation formulation
$\mathcal{H}$	Ratio coefficient relating the two rational degradation functions
$\mathcal{H}^+$	History field enforcing damage irreversibility
$\bar{Y}_t$	Effective tension–shear damage-driving quantity
$\bar{Y}_0$	Threshold for damage initiation
l	Length-scale parameter of the variational damage model
b	Length-scale parameter of the unified phase-field model
c_v, c_{Tp}, c_p	Scaling parameters of the variational damage, Tpfcm, and unified phase-field formulations, respectively
ξ, ξ_v	Parameters of the crack geometric function
p	Exponent governing the stress-softening law
a_1, a_2, a_3	Constitutive parameters governing the stress-softening response
G_f	Fracture energy
G_f^I, G_f^{II}	Mode-I and Mode-II fracture energies, respectively
f_t	Tensile strength
τ_{cr}	Critical shear strength
χ	Shear-to-tensile strength ratio, $\chi = \tau_{cr}/f_t$
Φ_n^+	Tensile strain energy density on the critical plane
Φ_s	Shear strain energy density on the critical plane
Φ_{rem}	Remaining strain energy density
k_{v0}	Initial slope of the stress-softening law
w_{v1}	Ultimate softening displacement
<i>Material and thermal parameters</i>	
E	Young's modulus
ν	Poisson's ratio
λ, μ	Lamé constants
α	Coefficient of thermal expansion
ρ	Mass density
c	Specific heat capacity
$k(d)$	Damage-dependent thermal conductivity
k_0	Thermal conductivity of the undamaged material

thermal stresses at the bonding interfaces. These thermally induced stresses are widely recognized as the primary driving force for interfacial crack initiation and propagation, ultimately leading to electrical opens, signal degradation, and mechanical failure of the packaging structure [1,3]. Among the various constituents, the IMC layer formed at the solder-pad interface through reactive diffusion is particularly vulnerable due to its intrinsic brittleness, low fracture toughness, and unfavorable CTE mismatch with adjacent materials, making it one of the weakest links in the solder joint reliability chain [4–6]. Consequently, the development of predictive simulation tools capable of capturing the thermo-mechanical fracture behavior of these multi-material interconnection structures is of significant importance for understanding failure mechanisms, optimizing packaging design, and improving long-term reliability [3].

The phase-field method has emerged as a powerful computational framework for simulating fracture in solids [7–9]. By introducing a continuous damage variable that regularizes the sharp crack topology into a finite-width diffuse band, the method eliminates the need for explicit crack tracking, re-meshing, or predefined crack paths. Crack initiation, propagation, branching, and coalescence arise naturally from the solution of coupled field equations, making it particularly attractive for complex multi-material systems where crack paths are not known *a priori* [7,10]. Among the various phase-field formulations, the unified phase-field theory proposed by Wu [11] has received considerable attention. By incorporating material strength into the energy degradation function, this theory effectively removes the sensitivity of numerical results to the regularization length scale—a well-documented limitation of earlier models such as AT1 and AT2 [12]. The resulting phase-field regularized cohesive zone model (PF-CZM) provides a length-scale insensitive framework that bridges variational fracture mechanics and cohesive zone modeling [13,14]. Notably, the phase-field method has been implemented in several commercial finite element platforms, including Abaqus [15], LS-DYNA [16] and COMSOL Multiphysics [17–19], broadening its accessibility for engineering applications. Building on this foundation, the phase-field method has been progressively extended to thermo-mechanical fracture problems [20]. Badnava et al. [21] developed an *h*-adaptive thermo-mechanical phase-field model, Mandal et al. [22] proposed a computationally efficient monolithic solver for fracture in thermo-elastic solids, and Su et al. [23] presented a thermo-mechanically coupled gradient damage approach with application to three-dimensional thermal fracture. In the authors' previous work, a thermo-mechanically coupled phase-field model incorporating mixed tensile-shear failure modes was formulated within the PF-CZM framework and applied to fracture analysis of TSV and TGV interconnect structures [24], as well as sintered silver solder layers with random porous microstructures [2]. These studies demonstrated the capability of the phase-field method in capturing complex crack patterns driven by CTE mismatch in electronic packaging applications.

In the context of electronic packaging, thermo-elasto-plastic phase-field frameworks combined with Gaussian random field-generated porous morphologies have been developed to simulate the mechanical and fracture behaviors of sintered nano-silver [25], and the effects of temperature and strain rate on crack propagation in random porous sintered nano-silver have also been investigated within the phase-field framework [26]. However, the computational cost associated with phase-field fracture simulations remains a significant challenge, particularly for thermo-mechanical coupling problems in electronic packaging. In the PF-CZM framework, accurately resolving the damage gradient within the localization band demands fine spatial discretization. When combined with the staggered solution algorithm — where the displacement-temperature and damage subproblems are solved alternately at each time increment — the iterative procedure often requires a large number of iterations to achieve convergence, especially during rapid damage evolution [2,24]. For electronic packaging applications, the computational burden is further amplified by two factors: (i) the characteristic dimensions of critical features such as IMC layers (thickness $\sim 5\text{--}15\ \mu\text{m}$) and solder joints (diameter $\sim 0.3\text{--}0.6\ \text{mm}$) necessitate extremely fine meshes to resolve both the microstructural geometry and the phase-field regularization length scale; and (ii) the prescribed transient thermal histories require a large number of time increments to resolve the coupled temperature, displacement, and damage fields. Consequently, a single thermo-mechanical fracture simulation of an electronic packaging structure can become prohibitively expensive, motivating the development of more computationally efficient phase-field formulations that retain the theoretical rigor and length-scale insensitivity of the PF-CZM.

To address these computational challenges, Duan and coauthors [27,28] proposed a length-scale insensitive phase-field model termed Tpfcm, which shares the same energy degradation function and crack geometric function with the variational damage model [29,30], while retaining the same form of crack surface density function as the PF-CZM. As a consequence, the Tpfcm inherits both the length-scale insensitivity of the PF-CZM and the improved constitutive self-consistency of the variational damage theory. Specifically, the constitutive parameters of the Tpfcm are derived from a first-order equilibrium equation of the 1D inviscid problem, avoiding the approximation required in the PF-CZM where the damage and stress fields at the center of the localization band are assumed to apply uniformly to all spatial points within the band [11]. This improvement leads to enhanced convergence robustness and reduced computational cost. Numerical benchmarks presented in [27] demonstrated that the Tpfcm yields results in excellent agreement with the PF-CZM while achieving significant computational savings. However, the Tpfcm has so far been formulated and validated only for purely mechanical fracture problems at ambient temperature. Its extension to thermo-mechanical coupling, which is relevant to fracture analysis of electronic packaging structures, has not yet been addressed.

The present work builds on two distinct lines of previous research. The authors' previous studies developed thermo-mechanically coupled PF-CZM formulations and employed a tension-shear energy decomposition for electronic-packaging fracture [2,24], whereas the original Tpfcm [27] was formulated for purely mechanical fracture. Relative to the original Tpfcm, the present contribution is to formulate and implement a thermo-mechanically coupled Tpfcm by incorporating thermal strain into its elastic response and damage-driving force, coupling damage to heat conduction through the Tpfcm degradation function, and adapting the existing tension-shear decomposition to its variational-damage-based constitutive structure. Relative to the previous thermo-mechanical PF-CZM models, the resulting framework retains the Tpfcm constitutive-parameter identification based on the first-order equilibrium equation of the one-dimensional inviscid problem rather than the PF-CZM localization-band approximation. Its fracture response and

numerical performance are assessed through direct comparisons with the PF-CZM, followed by applications to IMC layers, full-IMC solder joints, and TGV structures.

The remainder of this paper is organized as follows. Section 2 presents the Tpfczm phase-field framework. Section 3 presents numerical benchmark assessments of the Tpfczm, including comparisons with the PF-CZM and an evaluation of the damage-dependent thermal conductivity. Section 4 presents numerical examples that illustrate the capability of the proposed approach for electronic packaging applications. Finally, Section 5 summarizes the main findings and conclusions.

2. Tpfczm phase-field framework

The Tpfczm phase-field model [27] adopts the theoretical framework of the unified phase-field model [11] and incorporates key elements from variational damage theory [29]. The resulting formulation retains the length-scale insensitivity of the unified phase-field model while offering improved computational convergence. In this section, the thermo-mechanically coupled Tpfczm framework is systematically presented.

Let $\Omega \subset \mathbb{R}^{\delta}$ ($\delta \in \{1, 2, 3\}$) denote the reference configuration occupied by a material body. The body is subjected to displacement $\bar{\mathbf{u}}$, temperature $\bar{T}$, traction $\bar{\mathbf{t}}$, and heat flux $\bar{\mathbf{q}}$, which are imposed on the Dirichlet-type boundaries ($\partial\Omega_u, \partial\Omega_T$) and Neumann-type boundaries ($\partial\Omega_t, \partial\Omega_q$), respectively. The external boundary satisfies $\partial\Omega = \partial\Omega_u \cup \partial\Omega_t$ with $\partial\Omega_u \cap \partial\Omega_t = \emptyset$, and the thermal boundary satisfies $\partial\Omega = \partial\Omega_T \cup \partial\Omega_q$ with $\partial\Omega_T \cap \partial\Omega_q = \emptyset$. Let $\mathcal{B}$ denote the localization band around the sharp crack surface Γ inside the solid Ω . The objective is to analyze the crack evolution inside the solid Ω over the time domain $\mathcal{T} \subset \mathbb{R}$. A time-dependent damage field d is introduced to describe the localization band, which is defined as a mapping from the spatial domain $\mathcal{B}$ and the time domain $\mathcal{T}$ to the unit interval $[0, 1]$

$$d : \begin{cases} \mathcal{B} \times \mathcal{T} \rightarrow [0, 1] \\ (\mathbf{x}, t) \mapsto d(\mathbf{x}, t), \end{cases} \quad (1)$$

where $\mathbf{x} \in \mathcal{B}$ and $t \in \mathcal{T}$ denote the spatial and temporal coordinates, $d = 0$ indicates an intact material point, and $d = 1$ indicates a fully broken state.

2.1. Thermo-elastic constitutive relations

A temperature field $T : \Omega \times \mathcal{T} \rightarrow \mathbb{R}^+$ is introduced to describe the thermal state of the solid. For thermo-mechanical problems, the total strain tensor is additively decomposed into an elastic part ϵ^e and a thermal part ϵ^{th}

$$\epsilon = \epsilon^e + \epsilon^{th}. \quad (2)$$

For small-strain deformations, the displacement field is denoted by $\mathbf{u} := \mathbf{u}(\mathbf{x}, t)$, and the strain tensor is computed using the geometrically linear theory

$$\epsilon(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T). \quad (3)$$

The thermal strain tensor is defined as

$$\epsilon^{th} = \alpha(T - T_0) \mathbf{I}, \quad (4)$$

where α is the linear CTE, T_0 is the stress-free reference temperature, and $\mathbf{I}$ is the identity tensor. The elastic strain tensor is therefore

$$\epsilon^e = \epsilon - \epsilon^{th} = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) - \alpha(T - T_0) \mathbf{I}. \quad (5)$$

Therefore, the stored energy density of the damaged solid, considering isotropic degradation under thermo-mechanical coupling, is expressed in terms of elastic strain

$$\Phi_v(\epsilon^e, d) = \omega_v(d) \Phi_0(\epsilon^e), \quad (6)$$

where $\Phi_0(\epsilon^e)$ is the elastic strain energy density of the undamaged solid, and $\omega_v(d)$ is the energy degradation function of the variational damage theory.

$$\Phi_0(\epsilon^e) = \frac{\lambda}{2} [\text{tr}(\epsilon^e)]^2 + \mu \epsilon^e : \epsilon^e, \quad (7)$$

with λ and μ being the Lamé constants.

Accordingly, the Cauchy stress tensor of the damaged solid becomes

$$\sigma = \frac{\partial \Phi_v}{\partial \epsilon} = \omega_v(d) [\lambda \text{tr}(\epsilon^e) \mathbf{I} + 2\mu \epsilon^e] = \omega_v(d) \sigma_0(\epsilon^e), \quad (8)$$

where $\sigma_0(\epsilon^e)$ is the effective stress tensor evaluated at the elastic strain. The damage driving force under thermo-mechanical coupling is

$$\mathcal{Y}_v = -\frac{\partial \Phi_v}{\partial d} = -\omega'_v(d) \Phi_0(\epsilon^e). \quad (9)$$

It is important to note that the driving force (9) depends on the elastic strain energy $\Phi_0(\epsilon^e)$ rather than the total strain energy $\Phi_0(\epsilon)$. This distinction has a clear physical implication: a uniform and unconstrained thermal expansion, for which $\epsilon^e = \mathbf{0}$, does not contribute to damage evolution. Damage is driven exclusively by the constrained thermal stresses arising from CTE mismatch between adjacent materials or from mechanical boundary constraints. This is consistent with the well-established experimental observation that interfacial fracture in electronic packaging originates from CTE mismatch rather than from uniform heating [2,24].

2.2. Crack topology regularization

The sharp crack surface Γ is approximated by a regularized crack surface functional defined over the entire domain

$$\mathcal{A}(d) = \int_{\Omega} \gamma(d) dV, \quad (10)$$

where the crack surface density function of the Tpfcm is given by [27]

$$\gamma(d) = \frac{1}{c_v l} (\alpha_v(d) + l^2 |\nabla d|^2), \quad (11)$$

where $\alpha_v(d)$ is the crack geometric function of the variational damage theory, which will be detailed in Section 2.3, l is the length scale parameter, and ∇d is the gradient of the damage field. c_v is a scaling parameter introduced in the variational damage theory to ensure that the regularized crack surface functional equals the actual crack surface area, whose explicit expression is given by [27]

$$c_v = \frac{2b}{l} \cdot \frac{1}{\sqrt{\xi - \xi_v}} \left(\xi + \frac{1 - \xi_v}{2} \right), \quad (12)$$

where ξ and ξ_v are parameters of the crack geometric function and b is the length scale parameter of the unified phase-field model.

In this work, the length scale parameter of the variational damage model is set equal to that of the unified phase-field model, i.e., $l = b$. Under this condition, the crack surface density function (11) can be rewritten as

$$\gamma(d) = \frac{1}{c_{Tp} b} (\alpha_v(d) + b^2 |\nabla d|^2), \quad (13)$$

where the scaling parameter c_{Tp} is defined as

$$c_{Tp} = \frac{c_v l}{b}. \quad (14)$$

This formulation is formally identical to the crack surface density function of the unified phase-field model (PF-CZM) [11].

2.3. Crack geometric function and energy degradation function

The crack geometric function $\alpha_v(d)$ and the energy degradation function $\omega_v(d)$ are the two core building blocks of the phase-field model, governing the geometric description of the crack and the stiffness degradation behavior of the material, respectively. In the Tpfcm, both functions incorporate elements from the variational damage theory, which constitutes the key distinction from the classical PF-CZM.

The crack geometric function $\alpha_v(d)$ characterizes the mapping between the damage variable and the regularized crack surface density, and is defined as [27]

$$\alpha_v(d) = \xi d + (1 - \xi_v) d^2, \quad (15)$$

where $\xi \in [0, 2]$ originates from the unified phase-field theory and ξ_v ($\leq \xi$) is a coefficient introduced by the variational damage theory.

The energy degradation function $\omega_v(d)$ describes the degradation of strain energy during crack evolution. It is a monotonically decreasing function satisfying the following properties

$$\omega'_v(d) < 0, \quad \omega_v(0) = 1, \quad \omega_v(1) = 0, \quad \omega'_v(1) = 0. \quad (16)$$

The first condition ensures monotonic degradation; $\omega_v(0) = 1$ indicates that the intact material retains its full stiffness; $\omega_v(1) = 0$ ensures complete loss of load-carrying capacity at full damage; and $\omega'_v(1) = 0$ guarantees a smooth transition to the fully broken state.

In the Tpfcm framework, the degradation function is defined as [27]

$$\omega_v(d) := \frac{1}{1 + \phi_v(d)} = \frac{1}{1 + \mathcal{H}\phi_p(d)}, \quad (17)$$

where $\phi_v(d)$ and $\phi_p(d)$ are the rational degradation functions of the variational damage model and the unified phase-field model, respectively. The function $\phi_p(d)$ takes the form

$$\phi_p(d) = \frac{Q(d)}{(1 - d)^p}, \quad (18)$$

where the exponent $p > 0$ has a significant impact on the stress softening law. The two rational degradation functions are related through the ratio coefficient

$$\phi_v(d) = \mathcal{H}\phi_p(d), \quad \text{with} \quad \mathcal{H} = \frac{c_p b}{c_v l}, \quad (19)$$

where c_p is the scaling parameter of the unified phase-field model. The general expression of $\mathcal{H}$ is retained here to highlight the correspondence between the two models; in the present work with $l = b$, it reduces to $\mathcal{H} = c_p/c_v$. The coefficient $\mathcal{H}$ characterizes the relationship between the degradation functions of the two mechanical models. When $\mathcal{H} = 1$ (i.e., $c_p b = c_v l$), the degradation function (17) reduces to that of the unified phase-field model [11]

$$\omega_v(d) = \frac{(1-d)^p}{(1-d)^p + Q(d)}. \quad (20)$$

Furthermore, with appropriate choices of $Q(d)$ and p , the energy degradation function $\omega(d) = (1-d)^2$ of the commonly used classical phase-field model AT2 [10] can also be recovered, which demonstrates the unifying nature of the adopted framework.

The polynomial function $Q(d)$ in (18) is expressed as [11]

$$Q(d) = a_1 d + a_1 a_2 d^2 + a_1 a_2 a_3 d^3 = a_1 d \cdot P(d), \quad (21)$$

where $P(d) = 1 + a_2 d + a_2 a_3 d^2$, and a_1, a_2, a_3 are constitutive parameters governing the stress softening behavior. The determination of these parameters will be detailed in Section 2.4.

Together with the crack surface density function (13), which is formally identical to that of the PF-CZM, the energy degradation function $\omega_v(d)$ of the Tpfcm is also constructed to be consistent with the rational form employed in the PF-CZM. As a result, the critical stress $\sigma_c = f_t$ is determined solely by material parameters and remains independent of the length scale parameter b . Consequently, the Tpfcm inherits the same length-scale insensitivity as the PF-CZM, provided that the length scale is well resolved by the mesh [27].

2.4. Constitutive parameter identification

The constitutive parameters a_1, a_2 , and a_3 appearing in the polynomial function $Q(d)$ govern the stress softening behavior and must be determined prior to computation. Their identification follows a systematic procedure based on the 1D inviscid bar problem [11,27]: (i) a 1D bar model is established and the softening displacement expression is derived; (ii) an equivalent cohesive zone model relating the stress σ to the softening displacement w_v is constructed; (iii) the governing equations for the constitutive parameters are extracted from two limiting states, namely crack initiation ($d^* \rightarrow 0$) and complete failure ($d^* = 1$); and (iv) the resulting softening law is fitted to a classical softening curve to obtain the final parameter values.

Consider a sufficiently long bar occupying the 1D domain $x \in [-L, L]$, with an increasing displacement $\bar{u}$ applied at both ends. The crack initiates at the center $x = 0$, and the resulting localization band is confined to the interval $[-D_v, D_v]$ with $D_v < L$. At crack initiation ($d^* \rightarrow 0$), the failure strength f_t is obtained as [27]

$$f_t := \lim_{d^* \rightarrow 0} \sigma = \sqrt{\mathcal{A}_v \frac{\alpha'_v(0)}{\phi'_v(0)}}, \quad \mathcal{A}_v = \frac{2EG_f}{c_v l}, \quad (22)$$

where E is the initial elastic modulus of the material, G_f is the fracture energy, and $\alpha'_v(0)$ and $\phi'_v(0)$ denote the derivatives of the crack geometric function (15) and the rational degradation function (19) evaluated at $d = 0$, respectively.

The initial slope k_{v0} of the softening law is [27]

$$k_{v0} = -\frac{1}{8} c_v \frac{l}{b} \frac{f_t^2}{G_f} \frac{[\xi(a_2 + p + 1) - 1]^2}{\xi^2} \frac{1}{\sqrt{\xi - \xi_v}}, \quad (23)$$

and the ultimate softening displacement w_{v1} at complete failure ($d^* = 1$) is [27]

$$w_{v1} = \frac{4G_f b}{f_t c_v l} \sqrt{\frac{1 + \xi - \xi_v}{\xi - \xi_v}} \sqrt{\xi} \sqrt{P(1)} \lim_{d^* \rightarrow 0} (1 - d^*)^{1-p/2}. \quad (24)$$

From Eqs. (22)–(24), the three constitutive parameters can be determined. The parameter a_1 is obtained from the failure strength condition (22) as

$$a_1 = \frac{2E\xi}{c_p b f_t^2} = \frac{2\xi l_{ch}}{c_p b}, \quad \text{with} \quad l_{ch} = \frac{EG_f}{f_t^2}, \quad (25)$$

where l_{ch} is the Griffith characteristic length. The parameter a_2 is calculated from the initial slope condition (23) as

$$a_2 = \sqrt{\frac{-8k_{v0}G_f(\xi - \xi_v)}{(\xi + (1 - \xi_v)/2)f_t^2}} + \frac{1}{\xi} - (p + 1), \quad (26)$$

and a_3 is determined from the ultimate softening displacement condition (24) as

$$a_3 = \begin{cases} \frac{1}{a_2} \left(\frac{1}{\xi} \left(\frac{w_{v1} f_t (\xi + (1 - \xi_v)/2)}{2G_f} \right)^2 \frac{1}{1 + \xi - \xi_v} - (1 + a_2) \right), & p = 2, \\ 0, & p > 2. \end{cases} \quad (27)$$

It is worth noting that a_1 depends on the material properties (E , G_f , f_t) and the geometric function coefficients (ξ , ξ_v), but is independent of the softening law shape. The parameter a_2 is governed by the initial slope k_{v0} , while a_3 depends on both a_2 and the ultimate softening displacement w_{v1} . The softening law itself is determined solely by a_2 and a_3 [27].

To obtain specific parameter values, the equivalent softening law must be fitted to a classical softening curve. In this work, the hyperbolic softening law [11] is adopted, given by

$$\sigma(w) = f_t \left(1 + \frac{f_t}{G_f} w \right)^{-2}, \quad (28)$$

with an initial slope of

$$k_0 = -\frac{2f_t^2}{G_f}. \quad (29)$$

This softening law is selected for three reasons: (i) it adequately captures the quasi-brittle fracture behavior typical of electronic packaging materials such as IMC layers and solder alloys; (ii) it involves few parameters and is straightforward to calibrate for engineering applications; and (iii) it exhibits good compatibility with the variational damage framework.

For the hyperbolic softening law, the typical constitutive parameters are [27]

$$\xi = \frac{1}{5}, \quad \xi_v = -\frac{1}{4}, \quad p = 4, \quad (30)$$

from which the remaining parameters are calculated as

$$a_2 = 2.9542, \quad a_3 = 0. \quad (31)$$

The parameter a_1 depends on the specific material properties and is computed according to (25) for each material in the numerical examples. The values of a_2 and a_3 given in (31) are adopted throughout all numerical examples.

2.5. Heat conduction with damage-dependent conductivity

When cracks develop within a material, the fully opened crack surfaces act as thermal barriers that impede heat flux across the fractured region. To capture this physical phenomenon, a damage-dependent thermal conductivity is introduced into the heat conduction equation.

The temperature evolution in the solid is governed by

$$\rho c \dot{T} = \nabla \cdot (k(d) \nabla T) + Q_h \quad \text{in } \Omega, \quad (32)$$

where ρ is the mass density, c is the specific heat capacity, $\dot{T}$ denotes the time derivative of the temperature, and Q_h represents an internal heat source. The heat flux vector follows the Fourier law

$$\mathbf{q} = -k(d) \nabla T. \quad (33)$$

The degraded thermal conductivity $k(d)$ is defined as

$$k(d) = \omega_v(d) k_0, \quad (34)$$

where k_0 denotes the thermal conductivity of the intact material. The degradation function $\omega_v(d)$ used in the mechanical constitutive relation (6) is also applied to thermal conductivity as an engineering simplification. This choice uses the same phase-field variable to describe the loss of mechanical load-carrying capacity and the degradation of heat transport, while avoiding additional transport-specific phenomenological parameters. It satisfies $k(0) = k_0$ for the intact material and $k(1) = 0$ within the fully damaged region, the latter representing an idealized thermally insulating crack. Such a treatment has been adopted in thermo-mechanical phase-field models [2].

Eqs. (32) and (34) establish feedback from the damage field to the temperature field: as damage evolves, the effective thermal conductivity decreases and modifies the temperature distribution. Conversely, the temperature field affects the mechanical and damage responses through thermal strain and the resulting stress and damage-driving force. The present formulation is therefore described as a two-way coupled thermo-mechanical-damage framework.

2.6. Tension-shear energy decomposition

In electronic packaging applications, the dominant failure mechanism of IMC layers and solder joints is shear-induced interfacial cracking driven by CTE mismatch. Classical energy decomposition schemes — such as the volumetric-deviatoric decomposition [31] and the spectral decomposition [10] — do not explicitly separate the shear energy contribution and may therefore underestimate shear-induced damage. To address this limitation, a tension-shear energy decomposition based on the formulation of Wang et al. [32] is adopted. This decomposition has been successfully applied to electronic packaging problems in the authors' previous studies [2,24]; only the key equations are summarized here within the Tpfcm framework.

The elastic strain energy density is decomposed into three parts

$$\Phi_0(\epsilon^e) = \Phi_n^+(\epsilon^e) + \Phi_s(\epsilon^e) + \Phi_{\text{rem}}(\epsilon^e), \quad (35)$$

where Φ_n^+ is the tensile strain energy density induced by the normal tensile stress on a critical plane, Φ_s is the shear strain energy density on the same plane, and Φ_{rem} denotes the remaining strain energy. The tensile and shear energy contributions are degraded using the same degradation function $\omega_v(d)$, yielding the stored energy density of the damaged solid

$$\Phi_v(\epsilon^e, d) = \omega_v(d) [\Phi_n^+(\epsilon^e) + \Phi_s(\epsilon^e)] + \Phi_{\text{rem}}(\epsilon^e) \quad (36)$$

The fracture energies for pure tensile (mode I) and pure shear (mode II) failure are defined as

$$G_f^I = G_f, \quad G_f^{II} = \chi^2 \frac{E}{\mu} G_f, \quad (37)$$

where $\chi = \tau_{\text{cr}}/f_t$ is the ratio of the critical shear strength τ_{cr} to the tensile strength f_t . The parameter χ governs the dominant fracture mode: $\chi \geq \sqrt{2}/2$ corresponds to tension-dominated failure (maximum normal stress criterion), while $\chi < \sqrt{2}/2$ corresponds to shear-dominated failure (Mohr–Coulomb type), with the limiting case $\chi = 0$ reducing to a pure shear (Tresca) criterion. For detailed derivations, the reader is referred to [24,32].

For the plane stress state, let $\bar{\sigma} = \sigma_0(\epsilon^e)$ denote the effective (undamaged) stress tensor, and let $\bar{\sigma}_1$ be its first principal stress. The critical plane is defined as the plane on which the combined tensile and shear energy contribution is maximized; its unit normal and tangential vectors are denoted by $\mathbf{r}$ and $\mathbf{m}$, respectively.

The effective normal and shear stresses on this plane are

$$\bar{\sigma}_n = \mathbf{r} \cdot \bar{\sigma} \cdot \mathbf{r}, \quad \bar{\tau}_s = \mathbf{m} \cdot \bar{\sigma} \cdot \mathbf{r}. \quad (38)$$

The energy density components are then computed as

$$\Phi_n^+ = \frac{\langle \bar{\sigma}_n \rangle^2}{2E}, \quad (39)$$

$$\Phi_s = \frac{\bar{\tau}_s^2}{2\mu}, \quad (40)$$

$$\Phi_{\text{rem}} = \Phi_0(\epsilon^e) - \Phi_n^+ - \Phi_s, \quad (41)$$

where $\langle \cdot \rangle$ is the Macaulay bracket.

2.7. History field

To ensure the irreversibility of damage evolution ($\dot{d} \geq 0$) and to account for the fracture initiation threshold, a history field $\mathcal{H}^+$ is introduced. The effective driving energy is defined as a weighted sum of the tensile and shear strain energy densities [32]

$$\bar{Y}_t = \frac{\Phi_n^+(\epsilon^e)}{G_f^I} + \frac{\Phi_s(\epsilon^e)}{G_f^{II}}. \quad (42)$$

The energy threshold for damage initiation is

$$\bar{Y}_0 = \frac{f_t^2}{2EG_f}, \quad (43)$$

where f_t is the tensile strength. The history field is then defined as the maximum effective driving energy over the entire loading history

$$\mathcal{H}^+ = \max_{t \leq t} (\bar{Y}_0, \bar{Y}_t(t)). \quad (44)$$

This definition ensures that: (i) damage does not initiate until the effective driving energy exceeds the threshold $\bar{Y}_0$; (ii) the damage field is monotonically non-decreasing; and (iii) both the tensile and shear contributions are accounted for through their respective fracture energies.

2.8. Governing equations

The total potential energy of the damaged solid, obtained by combining the stored energy (6) and the regularized crack surface functional (10), is given by

$$\Pi(\mathbf{u}, d) = \int_{\Omega} \omega_v(d) \Phi_0(\epsilon^e) dV + G_f \int_{\Omega} \gamma(d) dV - \int_{\Omega} \mathbf{g} \cdot \mathbf{u} dV - \int_{\partial\Omega_t} \bar{\mathbf{t}} \cdot \mathbf{u} dA. \quad (45)$$

where $\mathbf{g}$ denotes the body force density. The heat conduction is treated as a separate evolution equation rather than included in the energy functional, consistent with the staggered solution scheme adopted in this work.

Taking the variation of (45) with respect to $\mathbf{u}$ and d , and combining it with the heat conduction Eq. (32), the tension-shear energy decomposition (36), and the history field (44), the governing equations of the thermo-mechanically coupled system in strong form read

$$\begin{cases} \nabla \cdot \boldsymbol{\sigma} + \mathbf{g} = \mathbf{0} & \text{in } \Omega, \\ \rho c \dot{T} = \nabla \cdot (k(d) \nabla T) + Q_h & \text{in } \Omega, \\ \omega'_v(d) \mathcal{H}^+ + \frac{\partial \gamma}{\partial d} = 0 & \text{in } \mathcal{B}, \end{cases} \quad (46)$$

where the stress tensor $\boldsymbol{\sigma}$ incorporates the tension-shear decomposition through Eq. (36), the degraded thermal conductivity $k(d)$ is defined by Eq. (34), and the variation of the crack surface density with respect to the damage field is

$$\frac{\partial \gamma}{\partial d} = \frac{1}{c_v l} [\alpha'_v(d) - 2l^2 \Delta d], \quad (47)$$

with $\Delta d = \nabla^2 d$ denoting the Laplacian of the damage field. In (46), the first equation governs the quasi-static mechanical equilibrium, the second describes the transient heat conduction, and the third controls the irreversible evolution of the damage field.

The boundary conditions for the coupled system are

$$\begin{cases} \boldsymbol{\sigma} \cdot \mathbf{n} = \bar{\mathbf{t}} & \text{on } \partial\Omega_t, \\ \mathbf{u} = \bar{\mathbf{u}} & \text{on } \partial\Omega_u, \\ k(d) \nabla T \cdot \mathbf{n} = \bar{q} & \text{on } \partial\Omega_q, \\ T = \bar{T} & \text{on } \partial\Omega_T, \\ \nabla d \cdot \mathbf{n}_B = 0 & \text{on } \partial\mathcal{B}, \end{cases} \quad (48)$$

where $\bar{q}$ is the prescribed heat flux on $\partial\Omega_q$. The initial conditions are

$$\mathbf{u}(\mathbf{x}, 0) = \mathbf{0}, \quad T(\mathbf{x}, 0) = T_0, \quad d(\mathbf{x}, 0) = 0 \quad \text{in } \Omega. \quad (49)$$

It should be noted that no artificial residual stiffness or residual thermal conductivity is used in the present formulation. Both properties are degraded directly by the Tpfcm degradation function and decrease to zero in the fully damaged state, representing the idealized complete loss of load-carrying and heat-conduction capacities within the damaged region. Although exact degradation to zero may, in principle, deteriorate the conditioning of the discrete system, the fully damaged zones in the present simulations remain localized within finite-width crack bands, while the surrounding intact or partially damaged material maintains sufficient connectivity and constraints for the global system. Accordingly, no singularity or convergence instability attributable to this treatment was observed in the numerical examples considered.

2.9. Finite element discretization and nonlinear solution procedure

To establish the finite element formulation of the coupled thermo-mechanical-damage problem, the mechanical equilibrium, heat conduction, and damage evolution equations in Eq. (46) are multiplied by the corresponding test functions and integrated over the computational domain. Applying integration by parts to the spatial gradient terms and incorporating the associated natural boundary conditions yields the following weak forms:

$$\begin{cases} \int_{\Omega} \boldsymbol{\sigma} : \nabla \delta \mathbf{u} dV - \int_{\Omega} \mathbf{g} \cdot \delta \mathbf{u} dV - \int_{\partial\Omega_t} \bar{\mathbf{t}} \cdot \delta \mathbf{u} dA = 0, \\ \int_{\Omega} \rho c \dot{T} \delta T dV + \int_{\Omega} k(d) \nabla T \cdot \nabla (\delta T) dV - \int_{\Omega} Q_h \delta T dV - \int_{\partial\Omega_q} \bar{q} \delta T dA = 0, \\ \int_{\Omega} \frac{2l}{c_v} \nabla d \cdot \nabla (\delta d) dV + \int_{\Omega} \left[\omega'_v(d) \mathcal{H}^+ + \frac{\alpha'_v(d)}{c_v l} \right] \delta d dV = 0. \end{cases} \quad (50)$$

The displacement, temperature, and damage fields are subsequently discretized in space using standard finite element shape functions. Following the Galerkin procedure, the discrete residual equations for the three fields are obtained, and the corresponding consistent tangent matrices are derived through linearization of the residuals. Details of the spatial discretization and element-level formulations can be found in the authors' previous works [2,24]. The present section therefore focuses on the solution procedure for the coupled three-field problem.

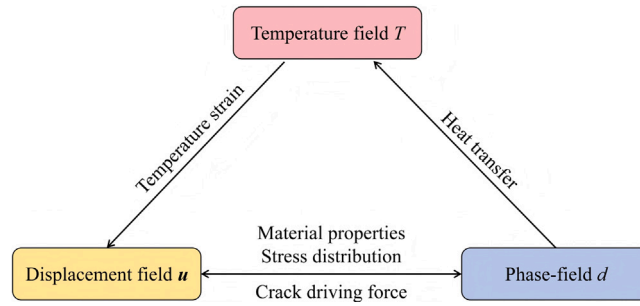


Fig. 1. Coupling relationships among the temperature, displacement, and damage fields in the thermo-mechanical Tpfcm framework.

Table 1

Comparison between the PF-CZM and Tpfcm formulations.

Feature	PF-CZM [11]	Tpfcm [27]
Crack geometric function	$\alpha_p(d) = \xi d + (1 - \xi)d^2$	$\alpha_v(d) = \xi d + (1 - \xi_v)d^2$
Degradation function	$\omega_p(d) = \frac{(1 - d)^p}{(1 - d)^p + Q(d)}$	$\omega_v(d) = \frac{1}{1 + H\phi_p(d)}$
Ratio coefficient	$H = 1$	$H = c_p b / (c_v l)$

The transient term of the temperature field is discretized using the implicit backward Euler scheme, and the loading history is divided into a sequence of time intervals $[t_n, t_{n+1}]$, with $\Delta t = t_{n+1} - t_n$. The displacement, temperature, and damage fields are mutually coupled, and the information transfer among the three fields is illustrated in Fig. 1. Based on these coupling relationships, a staggered iterative strategy is adopted, in which the three-field problem is divided into a thermo-mechanical subproblem and a damage subproblem that are solved alternately within each time increment.

The above staggered solution procedure is implemented through an Abaqus user element (UEL). At the element level, the UEL evaluates the thermal strain, stress, damage-driving force, degraded material stiffness, and damage-dependent thermal conductivity based on the current displacement, temperature, and damage variables, and constructs the corresponding element residual vectors and consistent tangent matrices. For each staggered subproblem, Abaqus/Standard assembles the element contributions into the corresponding global system and performs the Newton–Raphson iterations, during which the global residuals and solution corrections are evaluated against the convergence criteria. The solution at the current time increment is accepted only when the convergence requirements of both the thermo-mechanical and damage subproblems are satisfied after the latest staggered updates. The initial time increment is specified according to the loading history of each numerical example, while subsequent increments are adjusted automatically according to the convergence behavior. If convergence is not achieved, the current increment is repeated with a reduced increment size. The convergence tolerances, iteration limits, and automatic time-incrementation controls follow the default settings of Abaqus/Standard. The staggered partition is a numerical solution strategy and does not alter the two-way coupling relationships defined by the governing equations.

3. Numerical benchmark assessment

3.1. Comparison between Tpfcm and PF-CZM

The theoretical framework of the Tpfcm is systematically compared with the PF-CZM-based formulation employed in the authors' previous studies [2,24]. The variational damage theory adopted in the Tpfcm and the unified phase-field theory of the PF-CZM are connected through the following bridge relation between their crack geometric functions [27]

$$\alpha_v(d) = \alpha_p(d) + b^2 |\nabla d|^2, \quad (51)$$

where $\alpha_p(d) = \xi d + (1 - \xi)d^2$ is the crack geometric function of the PF-CZM. The core differences between the two formulations are summarized in Table 1.

A fundamental difference between the two formulations lies in the derivation of the softening displacement expression $w_v(d^*)$. In the PF-CZM, it is obtained from a second-order ordinary differential equation based on the 1D inviscid bar problem [11], which requires an approximation assuming that the damage and stress fields at the center of the localization band apply uniformly to all spatial points within the band [27]. In the Tpfcm, the variational damage theory provides a first-order governing equation, from which $w_v(d^*)$ can be derived directly through the bridge relation (51) without such approximation. This distinction contributes to the improved numerical performance of the Tpfcm, as will be demonstrated below.

To verify the numerical performance of the Tpfcm, a representative asymmetric notch benchmark is selected and compared against the PF-CZM under identical conditions. The same finite element mesh, material parameters, and boundary conditions are

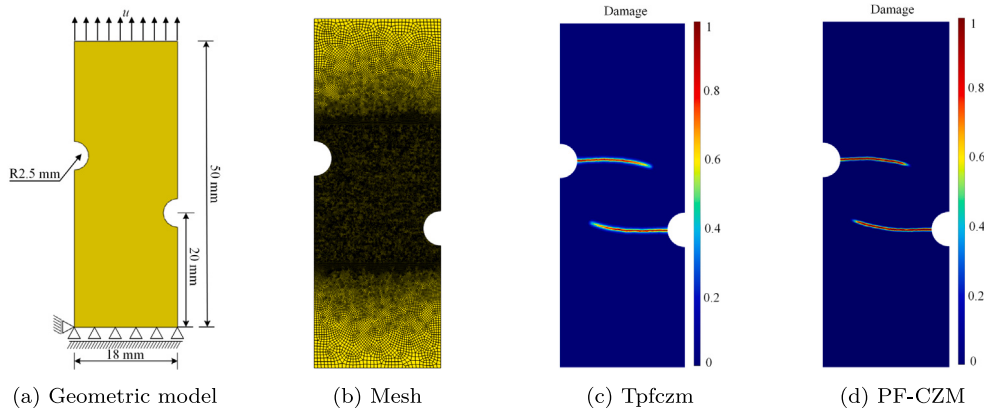


Fig. 2. Asymmetric-notch benchmark under a prescribed top-edge displacement: (a) geometry and boundary conditions, with the bottom edge constrained in the y -direction and the lower-left corner fixed in the x -direction; (b) mesh; (c) and (d) damage contours predicted by the Tpfczm and PF-CZM, respectively.

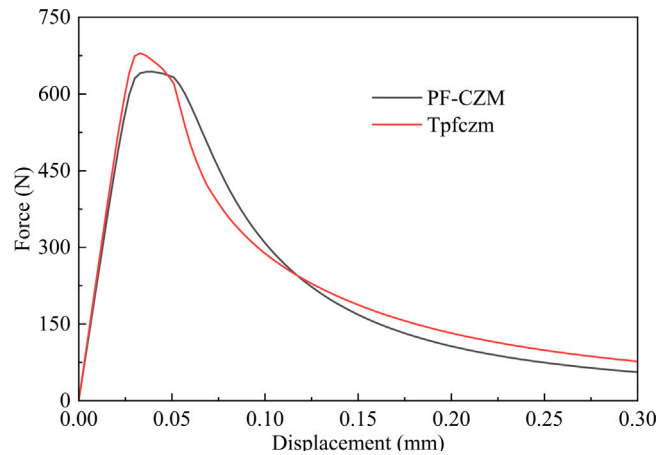


Fig. 3. Load-displacement curves predicted by the Tpfczm and PF-CZM for the asymmetric notch benchmark.

adopted for both methods. The geometric model and boundary conditions are shown in Fig. 2(a): the bottom boundary is constrained in the y -direction, the lower-left corner is fixed in the x -direction to prevent rigid body translation, and a prescribed displacement of 0.3 mm in the y -direction is applied to the top boundary. The model is discretized using four-node plane stress elements, with the mesh shown in Fig. 2(b).

Fig. 2(c) and (d) present the damage contours predicted by the two methods. In both models, the crack initiates from the notch tip and follows a similar overall propagation path, indicating qualitative consistency in the predicted fracture behavior. The load-displacement curves shown in Fig. 3 also exhibit comparable mechanical responses, with an approximately 5.35% difference in the predicted peak loads. The Tpfczm requires 128 iterations and 5.49 h, compared with 184 iterations and 7.75 h for the PF-CZM, corresponding to an approximately 29% reduction in computational time. Thus, for this benchmark, the Tpfczm provides a broadly comparable fracture response while reducing both the iteration count and computational time.

3.2. Assessment of damage-dependent thermal conductivity

To assess the influence of damage-dependent thermal conductivity on thermo-mechanically coupled fracture behavior, a simplified Cu-glass bilayer model is constructed, as shown in Fig. 4. The temperature prescribed on the upper boundary is linearly increased from 0 °C to 500 °C over a period of 500 min, while the lower boundary is maintained at 0 °C throughout the loading process, thereby generating heat conduction through the structure. To isolate the effect of conductivity degradation, two cases are considered. In the reference case, a constant thermal conductivity, $k = k_0$, is adopted and the influence of damage on heat conduction is neglected. In the damage-dependent case, the thermal conductivity proposed in the present framework, $k(d) = \omega_v(d)k_0$, is employed. Except for the treatment of thermal conductivity, the material properties, boundary conditions, and numerical settings are identical in the two cases.

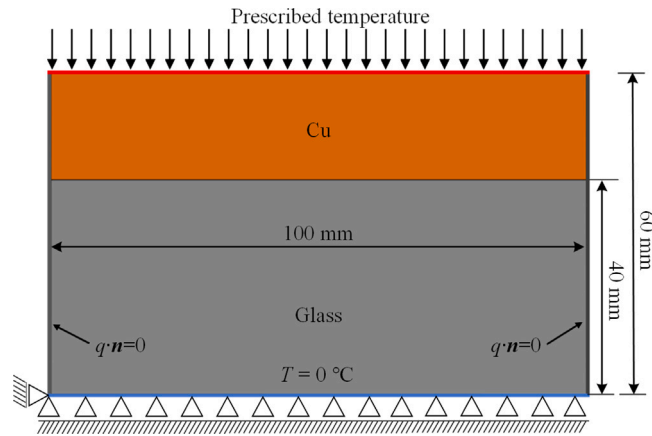


Fig. 4. Cu-glass bilayer model and thermal boundary conditions: the upper boundary is heated linearly from 0 to 500 °C over 500 min, while the lower boundary is maintained at 0 °C.

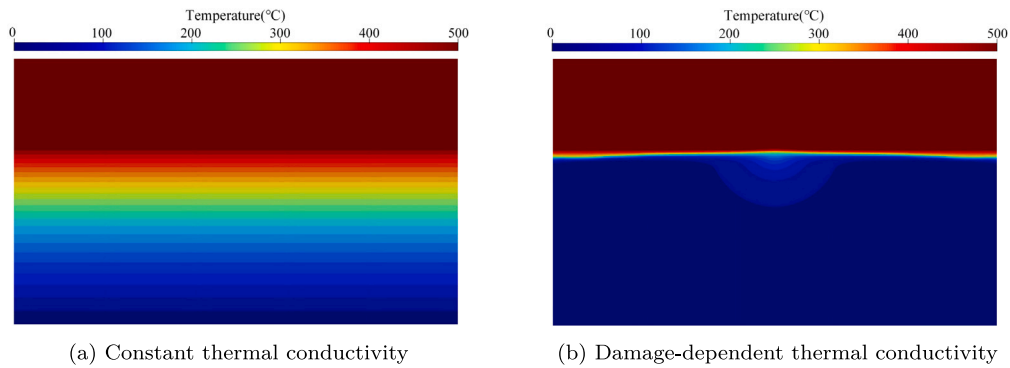


Fig. 5. Temperature distributions at $t = 500$ min obtained with constant and damage-dependent thermal conductivities.

The results at the end of the thermal loading process ($t = 500$ min) are compared in Fig. 5. In the constant-conductivity case, the temperature field varies relatively smoothly across the damaged region. In contrast, when the damage-dependent thermal conductivity is considered, the effective conductivity within the damaged zone progressively decreases as damage develops, resulting in a more pronounced local temperature drop and a steeper temperature gradient in the vicinity of the damage band. These results demonstrate that the damage-dependent conductivity formulation is capable of representing the thermal barrier effect associated with crack formation and that conductivity degradation can exert a noticeable influence on the evolution of the temperature field.

The corresponding damage fields are compared in Fig. 6. The two conductivity treatments predict essentially the same overall crack propagation path, whereas a slightly higher local damage level is observed when conductivity degradation is taken into account. This difference can be attributed to the reduction in thermal conductivity within the damaged region, which increases the local temperature gradient and consequently intensifies the thermal-strain incompatibility and associated thermo-mechanical stresses near the interface. The resulting redistribution of the local driving force slightly promotes subsequent damage evolution.

Taken together, Figs. 5 and 6 show that the adopted conductivity degradation produces a more pronounced local temperature drop and a steeper temperature gradient near the damaged region, while the overall crack path remains essentially unchanged and the local damage level is only slightly affected in this example. These results illustrate the numerical consequences of coupling thermal conductivity to the phase-field variable, but do not by themselves validate the specific relation $k(d) = \omega_v(d)k_0$. As discussed in Section 2.5, establishing a material-specific conductivity degradation law would require independent experimental measurements of crack-induced thermal transport degradation.

4. Numerical examples

Unless otherwise stated, all numerical examples are formulated as two-dimensional plane-stress problems. This idealization is used to examine the in-plane thermo-mechanical fracture behavior within a tractable computational framework, but it does not represent out-of-plane constraint, three-dimensional stress redistribution, or spatial crack-front evolution.

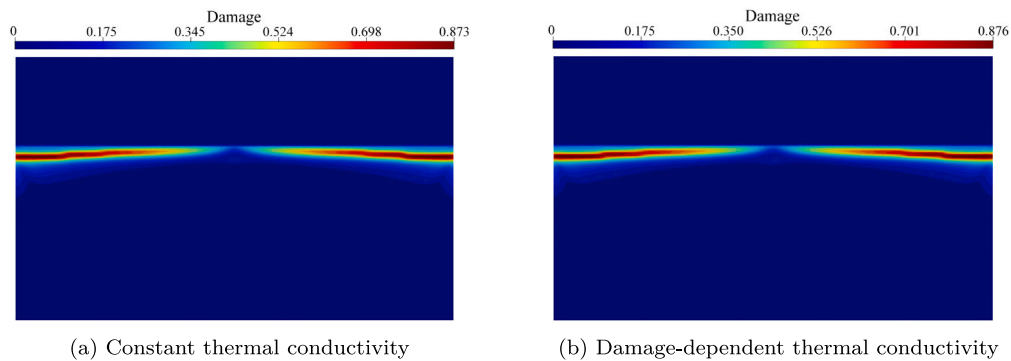


Fig. 6. Damage fields at $t = 500$ min obtained with constant and damage-dependent thermal conductivities.

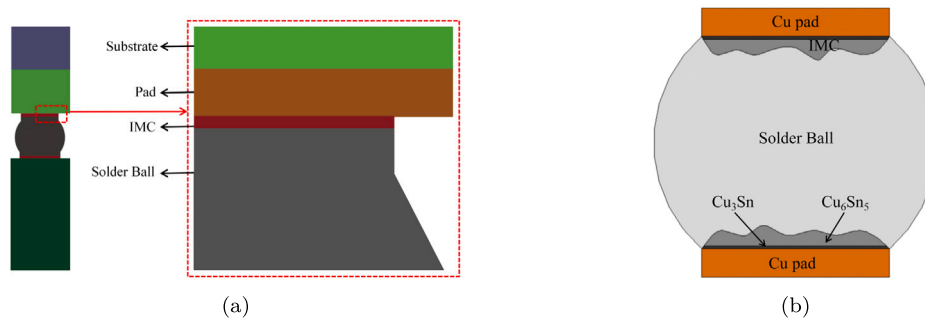


Fig. 7. (a) Location of the IMC layer at the solder ball-pad interface; (b) schematic illustration showing the Cu_3Sn and Cu_6Sn_5 sub-layers.

4.1. Shear fracture of the IMC layer with evolving phase composition

In electronic packaging structures, the IMC layer is an in-situ chemically bonded layer formed at the interface through reactive diffusion between the solder alloy and the metallic pad during soldering or reflow processes. This layer constitutes the physicochemical foundation for both electrical interconnection and mechanical bonding (Fig. 7(a)). The IMC layer is primarily composed of intermetallic compounds such as Cu_6Sn_5 and Cu_3Sn (Fig. 7(b)), which are characterized by a relatively high elastic modulus, extremely low fracture toughness, and a CTE that is mismatched with those of the adjacent materials. Under actual service conditions, its microstructural characteristics — including thickness, morphology (e.g., scallop-shaped structures), grain texture, and internal defects — undergo significant evolution with processing history and aging time, which in turn directly affects the mechanical performance and reliability. The evolution follows a well-established sequence: in the initial stage, Cu_6Sn_5 preferentially nucleates and grows on both sides of the Cu interface; as thermal aging progresses, Cu_3Sn gradually forms and expands between the Cu_6Sn_5 layer and the Cu substrate; in the final stage, the IMC layer is entirely replaced by Cu_3Sn .

In electronic packaging applications, damage in the IMC layer is primarily driven by shear stresses arising from CTE mismatch. Therefore, this study employs the Tpfcm phase-field method to investigate the fracture behavior of the IMC layer under shear loading. To characterize the influence of different phase compositions on the mechanical response, a Cu-IMC-Cu tri-layer model is constructed, and three representative configurations corresponding to distinct evolutionary stages of the IMC layer are considered (Fig. 8). The material parameters for each constituent phase are listed in Table 2, where only the IMC phases (Cu_6Sn_5 and Cu_3Sn) are considered to undergo damage, while the Cu substrate is treated as linear elastic. In all three configurations, the total thickness of the IMC layer is fixed at $30\ \mu\text{m}$. Specifically, Model I corresponds to the initial evolutionary stage (Fig. 8a), in which the IMC layer consists of two $3\ \mu\text{m}$ -thick Cu_3Sn sub-layers on either side of a $24\ \mu\text{m}$ -thick Cu_6Sn_5 core; Model II corresponds to an intermediate state (Fig. 8b), with two $10\ \mu\text{m}$ -thick Cu_3Sn sub-layers sandwiching a $10\ \mu\text{m}$ -thick Cu_6Sn_5 core; and Model III corresponds to the final state (Fig. 8c), in which the entire IMC layer is composed of Cu_3Sn . Note that the shear displacement is applied directly in this example to isolate the effect of phase composition on the fracture resistance, without introducing the additional complexity of a thermo-mechanical loading history.

Before comparing the damage evolution among the three configurations, the sensitivity of the numerical results to the mesh size and the length-scale parameter is first examined using Model II as a representative case. A shear displacement of $u = 1\ \mu\text{m}$ is applied to the top boundary (Fig. 8d).

A mesh convergence study is first conducted to evaluate the influence of mesh density on the numerical results. Four meshes containing $N = 13\,643$, $19\,870$, $33\,271$, and $69\,941$ elements are considered. As shown in Fig. 9, taking the result obtained with the

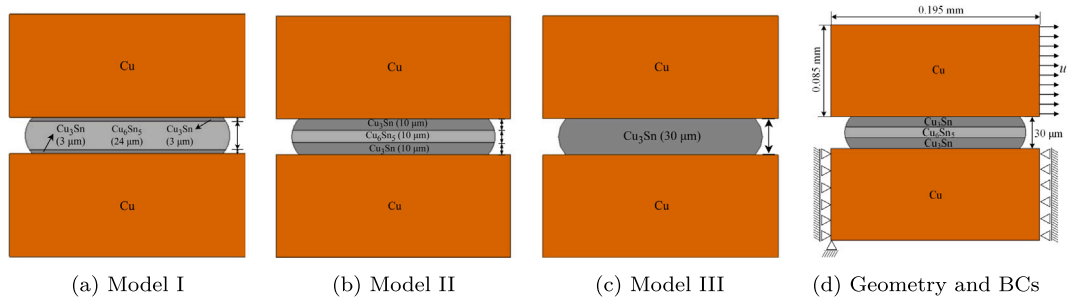


Fig. 8. Cu-IMC-Cu tri-layer model under a prescribed top-boundary shear displacement of $u = 1 \mu\text{m}$: (a)–(c) representative configurations with a dominant Cu_6Sn_5 core, comparable Cu_3Sn and Cu_6Sn_5 thicknesses, and pure Cu_3Sn , respectively; (d) geometry and displacement boundary conditions.

Table 2

Material parameters of the constituent phases in the IMC shear model. The Cu substrate is treated as linear elastic [33,34].

Material	E (MPa)	ν	f_t (MPa)	G_f (N/mm)
Cu_6Sn_5	1.260×10^5	0.30	50	0.0559
Cu_3Sn	1.342×10^5	0.33	75	0.0655
Cu	1.280×10^5	0.30	–	–

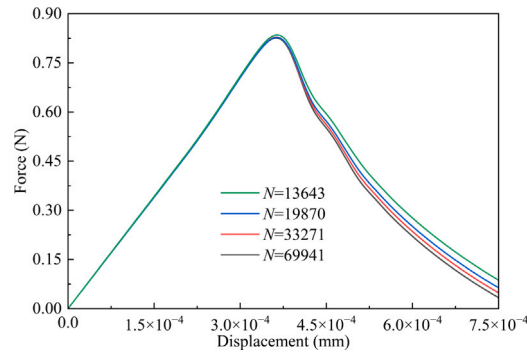


Fig. 9. Load-displacement curves under four different mesh densities.

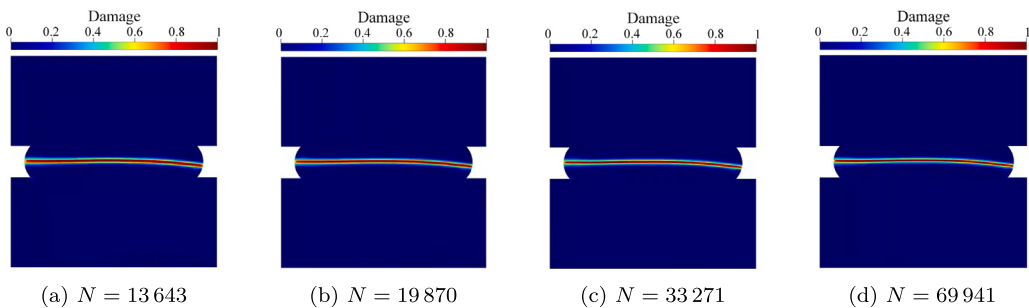


Fig. 10. Damage contours under four different mesh densities.

finest mesh of $N = 69941$ elements as the reference, the load-displacement curves obtained with $N = 19870$ and 33271 elements are already close to that of the finest mesh, with corresponding peak-load errors of 0.31% and 0.21%, respectively. In contrast, the load-displacement response obtained with $N = 13643$ elements exhibits a relatively larger deviation, with a peak-load error of 1.13%. In addition, the damage contours in Fig. 10 show that all four meshes consistently capture similar crack propagation paths and damage morphologies. Considering both computational accuracy and cost, the mesh with $N = 19870$ elements is adopted for the subsequent simulations.

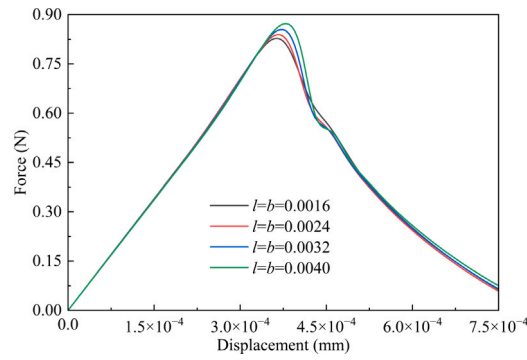


Fig. 11. Load-displacement curves under four different length-scale parameters.

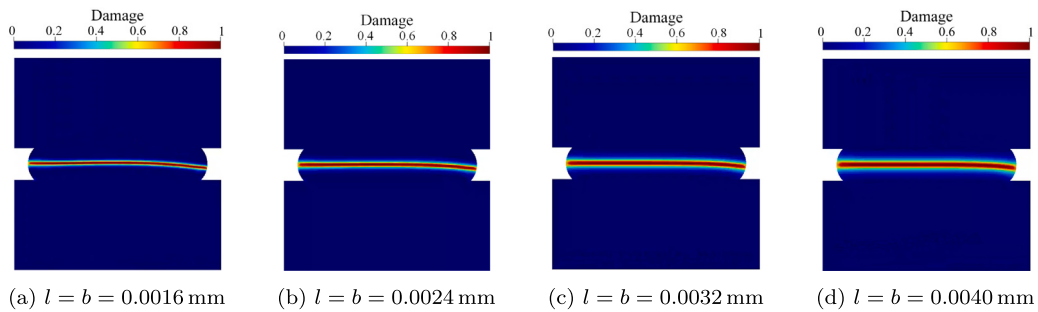


Fig. 12. Damage contours under four different length-scale parameters.

Subsequently, the influence of the length-scale parameter is investigated using the selected mesh with $N = 19870$ elements. The length-scale parameter is set to $l = b = 0.0016, 0.0024, 0.0032$, and 0.0040 mm, corresponding to 2, 3, 4, and 5 times the minimum element size, respectively. Since the length-scale parameter affects the width of the diffusive crack region, a certain variation in the load response is observed for different parameter values. As shown in Fig. 11, taking the result for $l = b = 0.0016$ mm as the reference, the deviations of the load-displacement curves for $l = b = 0.0024, 0.0032$, and 0.0040 mm are 1.50%, 1.65%, and 2.82%, respectively, while the overall trends remain similar. These results indicate that the influence of the length-scale parameter on the overall mechanical response is limited within the examined range. In addition, the damage contours in Fig. 12 exhibit similar crack propagation paths and fracture morphologies for the different length-scale parameters. Therefore, considering both crack resolution and computational efficiency, $l = b = 0.0016$ mm, i.e., twice the minimum element size, is adopted in the subsequent simulations.

The tensile strength, fracture energy, and CTE mismatch primarily govern crack initiation, post-initiation energy dissipation, and thermally induced stresses, respectively. A comprehensive sensitivity analysis involving these material-specific parameters across different constituent phases is beyond the scope of the present study. Because the present formulation focuses on mixed tensile-shear fracture, the shear-to-tensile strength ratio, defined in Section 2.6, is selected for a representative sensitivity analysis. Three values, $\chi = 0.1, 0.35$, and 0.6 , are considered.

As shown in Fig. 13, the load-displacement curves shift upward as χ increases. Taking $\chi = 0.6$ as the reference, the peak loads for $\chi = 0.1$ and 0.35 are lower by approximately 19.12% and 14.33%, respectively. Fig. 14 compares the damage fields at the same loading stage ($t = 50$ min). Although the overall crack paths remain similar, the local damage level increases as χ decreases.

This trend can be explained by the role of χ in regulating mixed-mode fracture. As the shear-to-tensile strength ratio decreases, the critical condition required for shear-dominated failure is reduced, thereby increasing the contribution of the shear component to damage evolution. Consequently, under the same loading condition, damage accumulates more rapidly and a higher local damage level is obtained. Considering the influence of χ on both the load-carrying response and damage evolution, and to maintain consistency in the parameter settings of the subsequent numerical examples, $\chi = 0.6$ is adopted throughout the following simulations.

Following the preceding mesh convergence study and the sensitivity analyses of the length-scale parameter and shear-to-tensile strength ratio, the damage evolution of the three configurations is analyzed under the same boundary conditions (Fig. 8d). The damage contours of the IMC layer for the three configurations are presented in Fig. 15. Significant differences in crack morphology can be observed as the phase composition evolves. In Model I (Fig. 15a), the crack initiates predominantly along the phase boundary between Cu_6Sn_5 and Cu_3Sn and subsequently extends into the Cu_6Sn_5 layer. In Model II (Fig. 15b), the crack propagates in a nearly horizontal manner within the intermediate Cu_6Sn_5 layer. In Model III (Fig. 15c), where the IMC layer has been entirely transformed into Cu_3Sn , the crack develops along the $\text{Cu}_3\text{Sn}/\text{Cu}$ interface and propagates inward. These differences in crack morphology and propagation path arise from the contrasts in the elastic and fracture properties of the constituent phases, which lead to spatial

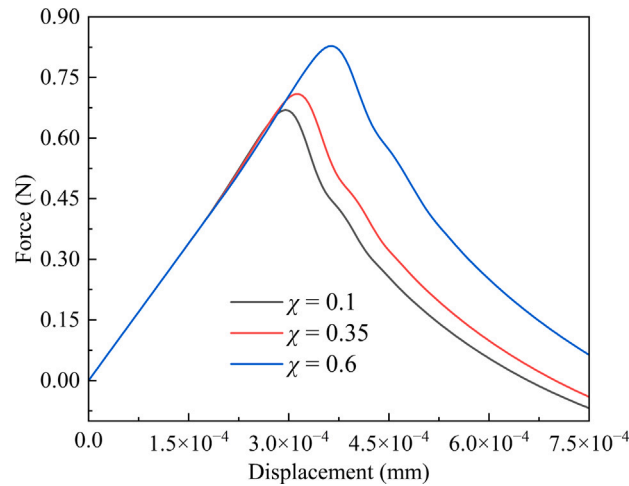


Fig. 13. Load-displacement curves obtained with different shear-to-tensile strength ratios.

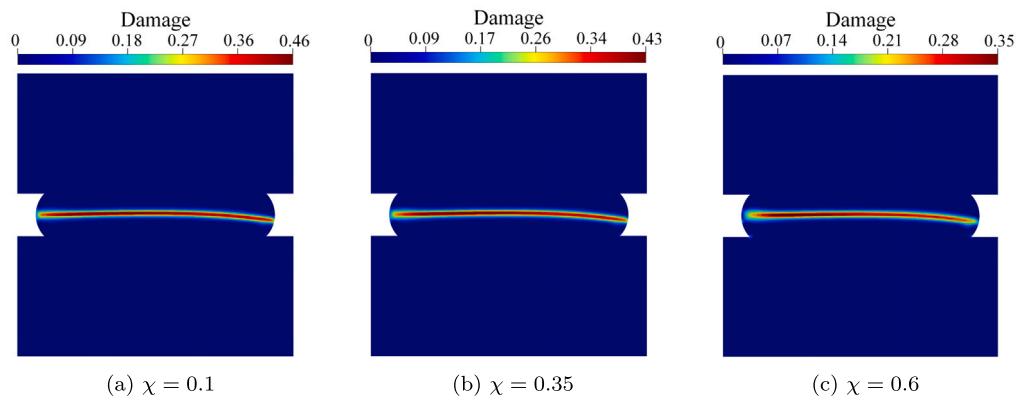


Fig. 14. Damage fields at $t = 50$ min obtained with different shear-to-tensile strength ratios.

variations in the local stress field and the effective mixed-mode damage-driving quantity $\bar{Y}_I$ near the material boundaries, while the damage-initiation threshold $\bar{Y}_0$ also differs between phases. Under the adopted parameter set, these effects favor damage development on the Cu_6Sn_5 side of the $\text{Cu}_6\text{Sn}_5/\text{Cu}_3\text{Sn}$ phase boundary. At the $\text{Cu}_3\text{Sn}/\text{Cu}$ boundary, the Cu substrate is modeled as linear elastic and is not allowed to undergo damage; consequently, damage localizes in the adjacent Cu_3Sn phase. Following damage initiation, stiffness degradation induces stress redistribution, which further guides crack propagation along the interface-adjacent regions. In the present conforming bulk phase-field discretization, material interfaces coincide with element boundaries, and the interface-adjacent elements retain the bulk constitutive and fracture parameters of their respective phases; no separate damage law or fracture parameters are assigned to the interface itself. Consequently, the jumps in material properties affect the local stress field, damage-driving force, and damage-initiation threshold. The local regularized crack path near an interface may also be influenced by mesh alignment and resolution. The predicted interface-adjacent paths should therefore be regarded as dependent on the adopted material parameters and discretization rather than as quantitative evidence of the intrinsic fracture resistance of the interfaces.

Fig. 16 presents the load-displacement curves of the IMC layer under shear loading for the three configurations. Significant differences in the peak load are observed among the three curves. Models I and II exhibit comparable peak loads, with Model II being slightly higher than Model I, whereas Model III attains a markedly higher peak load than the other two. This trend is closely related to the phase composition of the IMC layer in each configuration. In Models I and II, the primary fracture region remains within the Cu_6Sn_5 phase, where cracks initiate and propagate either within the bulk or along the phase boundaries, resulting in a relatively low overall load-carrying capacity and similar peak loads. The slightly higher peak load in Model II is attributed to the increased proportion of Cu_3Sn , which provides additional load-bearing capacity. In Model III, fracture occurs entirely within Cu_3Sn , which exhibits greater resistance to shear-induced damage under the adopted parameter set. Consequently, the peak load of the load-displacement curve is significantly higher than those of the other two configurations.

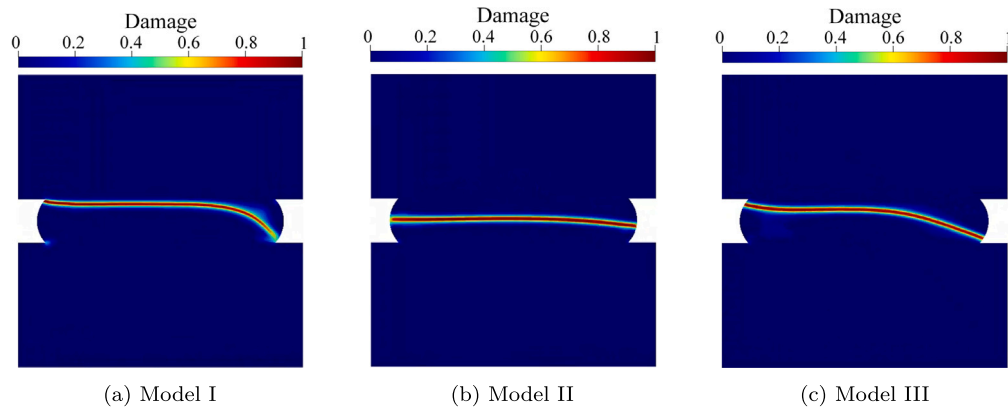


Fig. 15. Damage contours of the IMC layer under shear loading for three configurations.

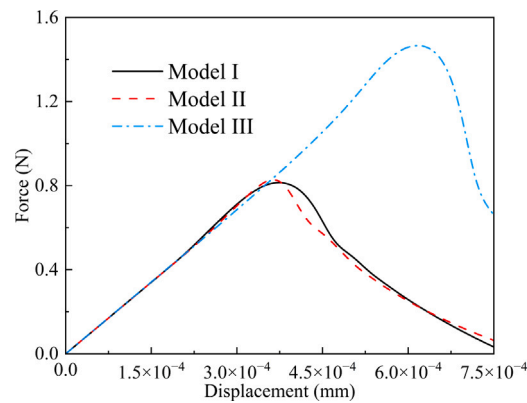


Fig. 16. Load–displacement curves of the IMC layer under shear loading for three configurations.

4.2. Thermo-mechanical fracture of the IMC layer with wavy morphology

Temperature changes generate CTE-mismatch-induced stresses at the IMC interfaces, which may drive crack initiation and propagation. In this example, the Tpfcmz is used to examine crack evolution under a single prescribed heating–holding–cooling history.

Fig. 17 illustrates the geometry, boundary conditions, and mesh discretization of the model. The solder ball has a diameter of 0.6 mm and a height of 0.5 mm. The copper layers at the top and bottom each have a thickness of 38 μm . The intermediate IMC layer, owing to its growth kinetics, exhibits a wavy morphology with an average thickness of approximately 12 μm . Regarding the boundary conditions, the vertical displacement along the bottom edge is constrained, and the lower-left corner node is fully fixed to prevent rigid body motion.

The prescribed temperature history is depicted in Fig. 18. The entire thermal loading process consists of three stages: a heating-up period from 0 to 60 min at a rate of 10 $^{\circ}\text{C}/\text{min}$, a holding period from 60 to 90 min during which the temperature is maintained at 600 $^{\circ}\text{C}$, and a cooling-down period from 90 to 150 min at a rate of -10 $^{\circ}\text{C}/\text{min}$.

The computational domain is discretized using four-node quadrilateral plane-stress elements. The IMC layer is meshed with a uniform element size of 0.001 mm, while the transition regions between the IMC layer and the solder ball employ a graded mesh with element sizes ranging from 0.001 to 0.01 mm. The material parameters adopted for each constituent are summarized in Table 3. It should be emphasized that all mechanical, fracture, and thermal material properties are treated as temperature-independent, while solder plasticity or viscoplasticity, creep, melting, and solidification are not considered. In particular, the prescribed peak temperature of 600 $^{\circ}\text{C}$ exceeds the melting range of SAC305; therefore, the calculated high-temperature response is not intended to represent the actual physical state of the solder. The prescribed thermal history is used as an accelerated numerical loading condition to illustrate CTE-mismatch-driven fracture mechanisms and crack-evolution trends rather than to provide quantitative predictions of the in-service performance of electronic packaging structures. To enable the observation of the complete fracture process within a tractable computational framework, the tensile strength and fracture energy of the IMC layer are deliberately scaled. The adopted values are $f_t = 50$ MPa and $G_f = 0.00559$ N/mm, corresponding to reductions by factors of approximately 20 and 10, respectively.

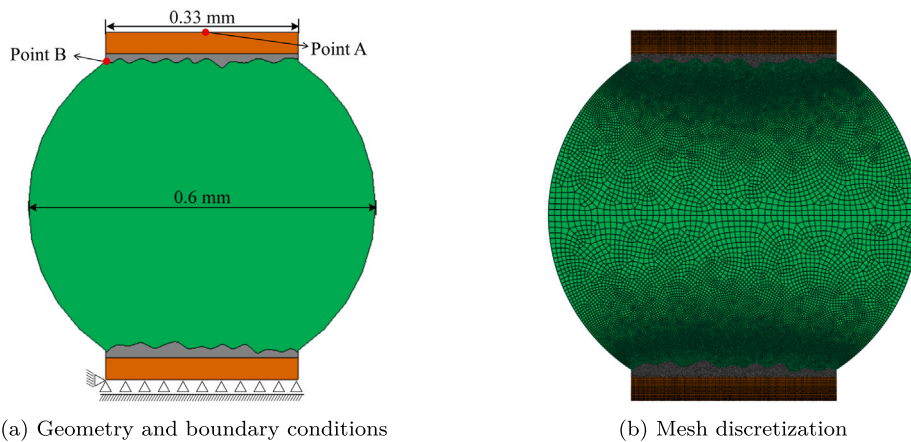


Fig. 17. Wavy-IMC solder-joint model: (a) geometry and displacement boundary conditions of the Cu-pad/IMC/solder configuration, with the bottom edge constrained vertically and the lower-left corner fixed; (b) mesh discretization.

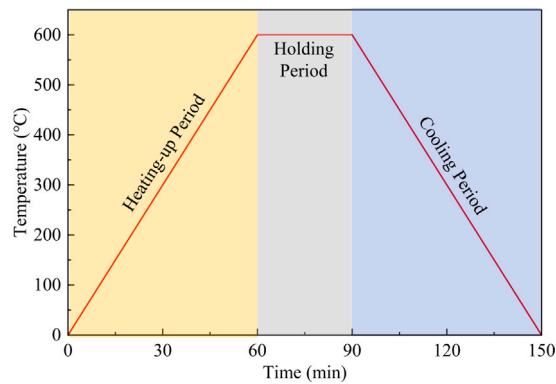


Fig. 18. Prescribed thermal history for the wavy-IMC model: heating to 600 °C over 60 min, holding until 90 min, and cooling to 0 °C by 150 min.

Table 3

Material parameters of the constituent layers in the IMC model [33,34].

Material	E (MPa)	ν	α ($^{\circ}\text{C}$)	f_t (MPa)	G_f (N/mm)
IMC (Cu_6Sn_5)	1.26×10^5	0.30	1.9×10^{-5}	50	0.00559
Cu pad	1.28×10^5	0.30	1.7×10^{-5}	–	–
Solder	3.35×10^4	0.35	2.5×10^{-5}	75	0.15

Having established the model, the stress distributions and damage evolution under thermo-mechanical coupling are analyzed. Fig. 19 presents the stress components and the damage field of the IMC layer model at the crack initiation stage ($t = 30.25$ min). The stress analysis reveals that the normal stress σ_{11} is predominantly concentrated in the central region of the IMC layer, particularly on the side adjacent to the solder ball. The shear stress σ_{12} is localized at the lateral edges where the solder ball meets the IMC layer, constituting the primary driving force for damage initiation. The normal stress σ_{22} is mainly distributed in the upper portion of the IMC layer; its contribution, combined with the shear stress, jointly promotes the damage evolution. The damage contour indicates that the damaged zone originates near the trough positions of the wavy interface at the corners where the solder ball meets the IMC layer. Combined with the stress analysis, it can be concluded that the shear stress σ_{12} is the dominant factor responsible for this damage initiation pattern.

Since the damage is most pronounced in the lower IMC layer, the following analysis focuses on the crack propagation behavior within this region. The predicted crack propagation paths are shown in Fig. 20. The damage first initiates at the trough positions of the wavy interface between the solder ball and the IMC layer, subsequently propagates from both lateral edges toward the center, and continues to advance along the undulating interface until near-complete penetration is achieved. This result indicates that the dominant fracture mode is interfacial failure between the solder ball and the IMC layer.

A qualitative comparison between the numerical prediction and the experimental observations reported by Mo et al. [35] is presented in Fig. 21. Both the numerical and experimental results exhibit interface-dominated crack propagation with a characteristic

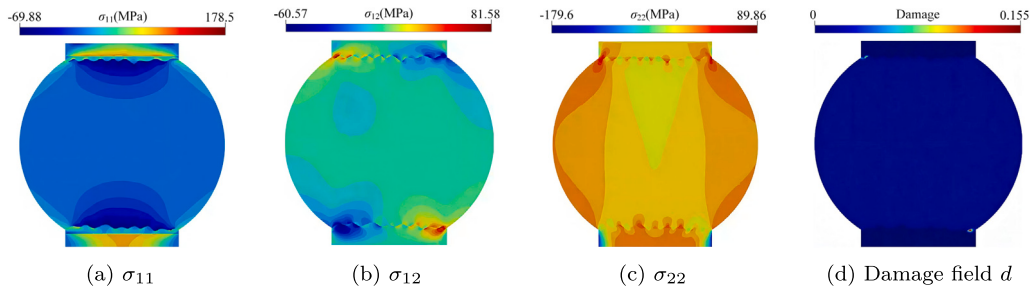


Fig. 19. Stress distributions and damage field at the crack initiation stage ($t = 30.25$ min).

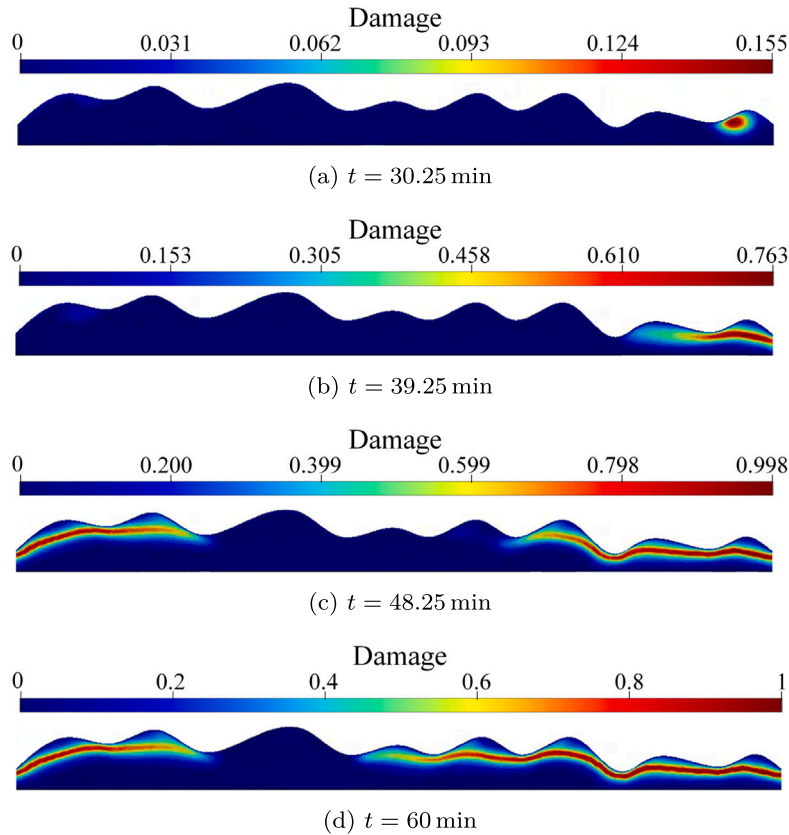


Fig. 20. Crack propagation path in the lower IMC layer at four time instants.

wavy morphology along the IMC layer. Although this qualitative consistency supports the physical plausibility of the predicted fracture pattern, the experimental configuration and loading conditions do not exactly match those of the present numerical model. Therefore, the comparison should be interpreted as qualitative morphological corroboration rather than quantitative validation.

To further examine the influence of the wavy IMC morphology on local fracture behavior, an additional flat-interface reference model is established. The material parameters, boundary conditions, and thermal loading are kept unchanged, and the two models are compared in terms of shear-stress localization, damage-initiation location, and final crack path.

Fig. 22 compares the shear stress distributions near the left edge of the IMC layer in the wavy-interface and flat-interface models at the same loading instant ($t = 30.25$ min). Both models exhibit pronounced shear-stress localization near the lateral edge of the solder/IMC interface, which is consistent with the important role of shear deformation in damage initiation. In the flat-interface model, the high-stress region is concentrated near the geometric corner, whereas in the wavy-interface model it shifts toward the adjacent valleys. Consistent with this redistribution, damage in the wavy-interface model initiates preferentially near the valley regions. These results indicate that the wavy morphology modifies the spatial location of local stress concentration and, consequently, the preferred sites of damage initiation.

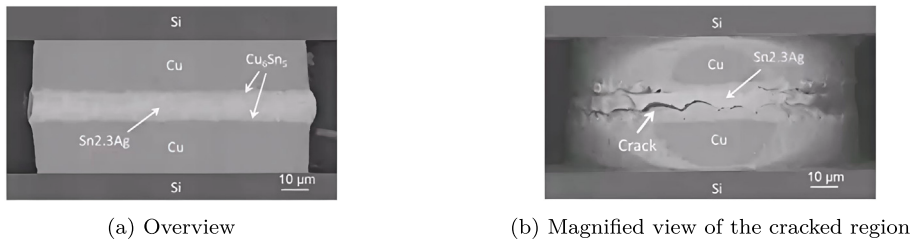


Fig. 21. Experimental observation of IMC layer fracture from Ref. [35].

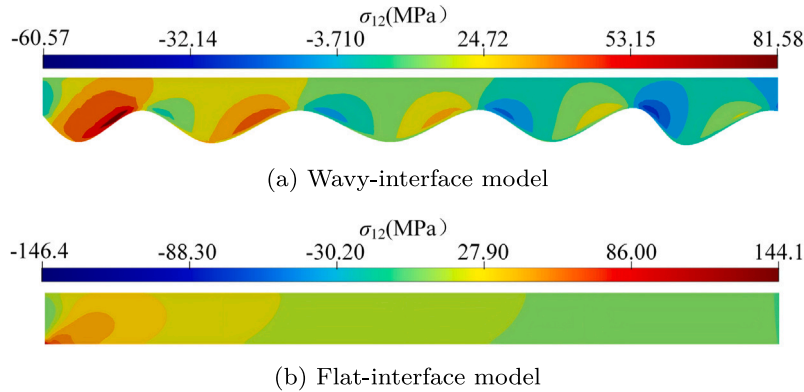


Fig. 22. Shear-stress distributions in the wavy- and flat-interface IMC models at the same loading instant ($t = 30.25$ min) under identical material properties, boundary conditions, and thermal loading.

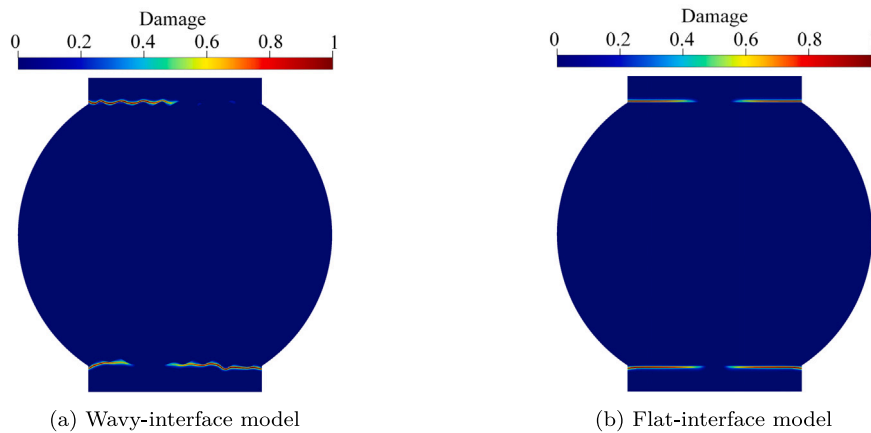


Fig. 23. Final crack propagation patterns of the wavy-interface and flat-interface IMC models.

Fig. 23 compares the final crack patterns predicted by the two models. In both cases, cracks propagate from the two lateral sides toward the central region along the solder/IMC interface, indicating the same overall interface-dominated failure mode. However, the flat-interface model produces approximately straight and nearly symmetric crack paths, whereas the cracks in the wavy-interface model undergo repeated local deflection following the undulating interface and exhibit a certain degree of asymmetry. Therefore, the wavy morphology affects the local stress localization, damage-initiation sites, and crack paths without changing the overall failure mode.

4.3. Thermo-mechanical fracture of a full-IMC solder joint

In recent years, full-IMC solder joints have gained increasing attention in electronic packaging applications owing to their high thermal resistance, excellent thermal stability, superior mechanical properties and creep resistance, and strong interfacial bonding

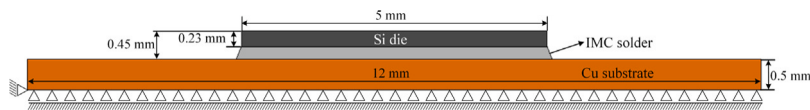


Fig. 24. Geometry and displacement boundary conditions of the model subjected to a uniform temperature increase.

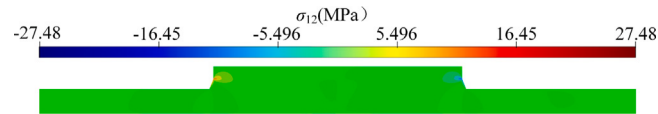


Fig. 25. Shear stress (σ_{12}) distribution at the crack initiation stage.

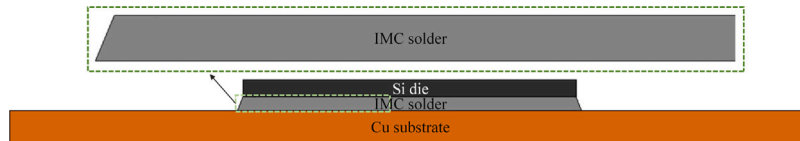


Fig. 26. Magnified view of the left portion of the model, indicating the region of interest for detailed crack propagation analysis.

Table 4

Characteristic temperatures predicted by the Tpfczm and PF-CZM.

Model	Crack initiation temperature	Failure temperature
PF-CZM	111.6 °C	131.3 °C
Tpfczm	107.8 °C	144.4 °C

strength, making them particularly suitable for microelectronic and advanced packaging technologies. To investigate the fracture behavior of full-IMC solder joints under thermo-mechanical coupling, a two-dimensional Si die–full IMC–Cu substrate tri-layer model is established, and the Tpfczm phase-field method is employed to simulate the damage evolution process. A uniform temperature increment of $\Delta T = 300$ °C is prescribed as an idealized thermal loading condition for the numerical demonstration. The displacement boundary conditions are illustrated in Fig. 24. The full-IMC layer is treated as Cu_6Sn_5 , and its material parameters are identical to those listed in Table 3.

Fig. 25 shows the shear stress distribution at the crack initiation stage. Due to the CTE mismatch between the constituent materials, stress concentration first develops at the corner regions of the Si die/IMC interface. To examine the crack initiation and propagation behavior in greater detail, the left portion of the IMC layer, as indicated in Fig. 26, is selected for magnified analysis.

Fig. 27 presents four representative stages of crack propagation within the selected region. The crack initiates at the Si die/IMC interface corner and subsequently propagates in a stable manner along the interface under thermo-mechanical coupling, exhibiting a distinct interfacial failure mode. The final crack path across the entire model is shown in Fig. 28.

To compare the Tpfczm and PF-CZM under thermo-mechanical loading, the full-IMC solder-joint problem is analyzed using both formulations with identical geometry, prescribed material and fracture parameters, boundary conditions, and thermal loading. The formulation-specific constitutive coefficients are derived from the same material inputs according to the definitions of the respective models. Fig. 29 compares the damage distributions at $t = 33$ min, representing an early crack-propagation stage, whereas Fig. 30 presents a subsequent propagation stage at $t = 52.75$ min. In both models, damage is initially concentrated near the Si die/IMC interfacial corner and subsequently propagates inward along the interface. Although the predicted local crack paths do not completely coincide, both models predict the same dominant interfacial failure mode.

For quantitative comparison, crack initiation and failure are identified when the maximum damage variable reaches $d_{\max} = 0.4$ and $d_{\max} = 0.95$, respectively. As summarized in Table 4, the initiation temperatures predicted by the Tpfczm and PF-CZM are 107.8 °C and 111.6 °C, respectively. Thus, the Tpfczm predicts an initiation temperature 3.8 °C lower than that obtained with the PF-CZM. The corresponding failure temperatures are 144.4 °C and 131.3 °C, with the Tpfczm prediction being 13.1 °C higher. Overall, the two models show qualitative agreement in the initial damage region and dominant failure mode, while quantitative differences remain in the local crack evolution and predicted failure temperature.

Furthermore, void defects are commonly formed in the solder layer during the reflow process. These voids alter the local stress distribution and affect the crack propagation behavior, constituting a critical factor that compromises the reliability of electronic packaging components. To examine the influence of voids on the thermo-mechanical fracture response of the full-IMC layer, representative voids are introduced in the upper-left region of the IMC layer based on the defect-free model described above. Except for the inclusion of voids, all geometric dimensions, material parameters, and boundary conditions remain unchanged. The geometric configuration and void locations are illustrated in Fig. 31. By comparing the predictions with those of the defect-free model, the influence of voids on the predicted crack evolution is examined.

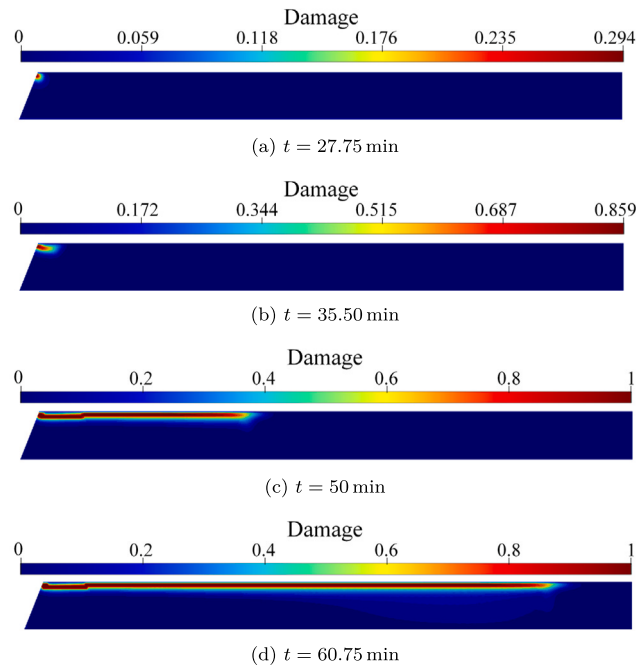


Fig. 27. Crack propagation in the magnified left region of the full-IMC model at four representative time instants.

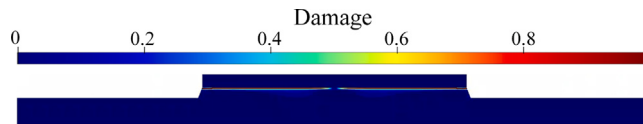


Fig. 28. Final crack path in the full-IMC solder joint model.

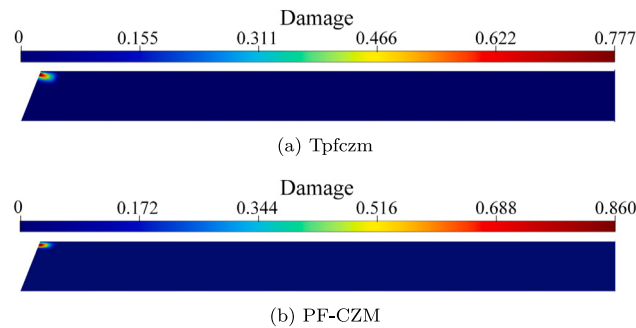


Fig. 29. Damage distributions predicted by the Tpfczm and PF-CZM at an early crack-propagation stage ($t = 33$ min).

Fig. 32 compares the shear stress (σ_{12}) histories at the crack initiation stage (0–30 min) between the defect-free and void-containing models. The shear stress in the void-containing model is notably higher than that in the defect-free model, with the maximum shear stress increasing by approximately 27%. This demonstrates that the voids significantly intensify the local stress concentration in the vicinity of the interface, which in turn directly affects the subsequent crack propagation behavior. To further quantify the influence of void defects on the thermo-mechanically coupled fracture process, Table 5 summarizes the crack initiation and failure temperatures of the defect-free and void-containing models. Using the same maximum-damage criteria for crack initiation and failure as defined above, the crack initiation temperature increases from 107.8 °C in the defect-free model to 117.2 °C in the void-containing model, indicating a certain difference between the two cases. In contrast, the failure temperature decreases from 144.4 °C to 142.5 °C. These results indicate that, although the void defects do not significantly alter the overall evolution trend during

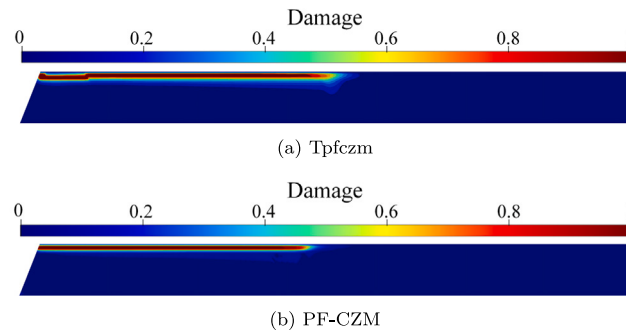


Fig. 30. Damage distributions predicted by the Tpfczm and PF-CZM at a representative crack propagation stage ($t = 52.75$ min).



Fig. 31. Representative void locations in the upper-left region of the Cu_6Sn_5 full-IMC layer; all other geometry, material properties, boundary conditions, and thermal loading are unchanged from the defect-free model.

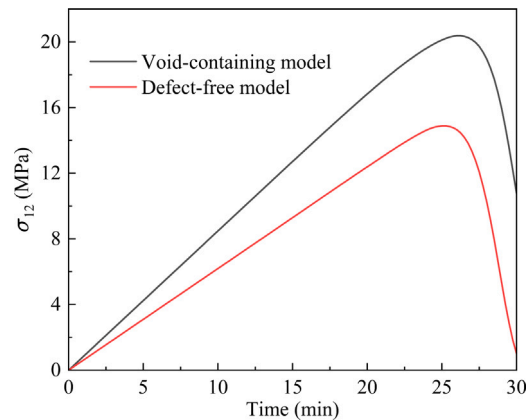


Fig. 32. Comparison of shear stress (σ_{12}) histories at the crack initiation stage between the defect-free and void-containing models.

Table 5

Crack initiation and failure temperatures of the defect-free and void-containing full-IMC solder joint models.

Model	Crack initiation temperature	Failure temperature
Defect-free model	107.8 °C	144.4 °C
Void-containing model	117.2 °C	142.5 °C

the crack initiation stage, they accelerate the subsequent damage accumulation process and cause the structure to reach the failure state earlier.

Figs. 33 and 34 show the local crack propagation paths in the left and right portions of the IMC layer, respectively, and Fig. 35 presents the final crack pattern of the entire model. The results reveal that the presence of voids significantly alters the crack propagation path. On the left side, the crack traverses the void region and continues to propagate along the Si die/IMC interface. On the right side, however, the crack path is deflected due to the stress redistribution induced by the voids on the left, causing the crack to deviate from the interface and extend into the interior of the IMC layer. These findings indicate that voids not only guide crack growth along their locations but may also promote crack penetration into the bulk of the IMC layer, thereby altering the overall fracture mode from purely interfacial delamination to a mixed interfacial–bulk failure.

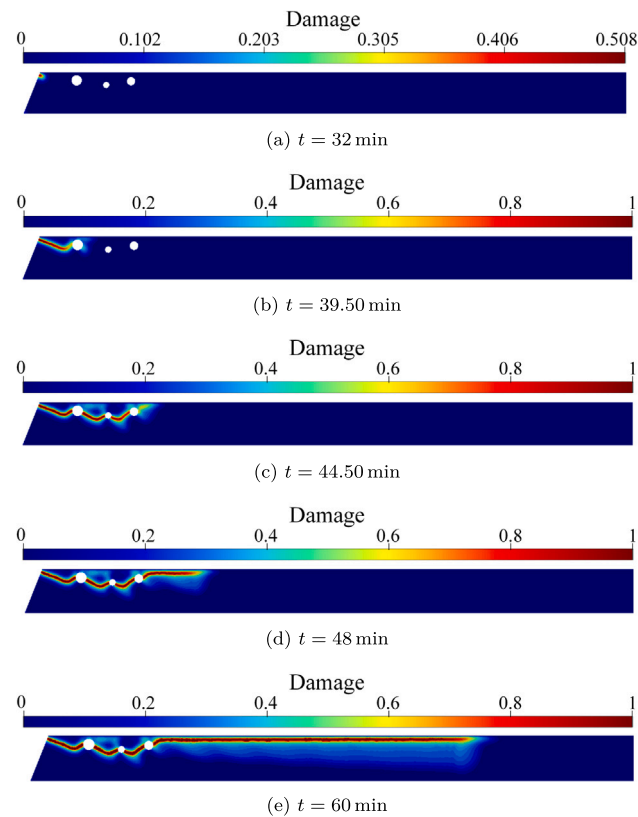


Fig. 33. Crack propagation path in the left portion of the IMC layer with voids.

Table 6

Material parameters of the constituent layers in the TGV model [24].

Material	E (MPa)	ν	α ($^{\circ}\text{C}$)	f_t (MPa)	G_f (N/mm)
Glass	7.78×10^4	0.17	4.8×10^{-7}	50	0.008
TGV copper	1.28×10^5	0.30	1.7×10^{-5}	—	—
Low- κ resin	2.45×10^3	0.35	6.6×10^{-5}	—	—

4.4. Thermo-mechanical fracture of through-glass vias

Through-glass vias (TGVs) serve as a critical interconnection structure in advanced 2.5D/3D packaging architectures. Due to the significant CTE mismatch between the TGV copper fill and the glass substrate, TGVs are prone to becoming the weakest link in the packaging structure under thermo-mechanical loading.

Here, the proposed method is employed to investigate the damage evolution behavior of TGVs under thermo-mechanical coupling. Compared with the PF-CZM-based analysis reported in the authors' previous work [24], the present study adopts the Tpfcm with hyperbolic softening law and a different TGV geometry incorporating a low- κ dielectric layer. The geometric model and displacement boundary conditions are illustrated in Fig. 36. The model comprises three constituents: TGV copper, the glass substrate, and a low- κ dielectric layer (resin material). The material parameters adopted for each constituent are summarized in Table 6. Regarding boundary conditions, the horizontal displacement is constrained on both the left and right edges, and the vertical displacement is fixed at the lower-left corner node to prevent rigid body motion. The prescribed temperature history is shown in Fig. 37. The thermal loading process consists of three stages: a heating-up period from 0 to 40 min at a rate of $15^{\circ}\text{C}/\text{min}$, a holding period from 40 to 65 min during which the temperature is maintained at 600°C , and a cooling-down period from 65 to 95 min at a rate of $-20^{\circ}\text{C}/\text{min}$.

To analyze the displacement evolution of the copper fills within the TGVs and elucidate the underlying damage mechanisms, the displacement fields at $t = 14.25$ min are extracted, as shown in Fig. 38. For clarity, the deformed configurations are displayed with a magnification factor of 20. In the horizontal direction (Fig. 38a), the lateral displacement of the copper fills is substantially larger than that of the glass substrate, indicating a pronounced tendency for relative sliding at the interface, which is the primary

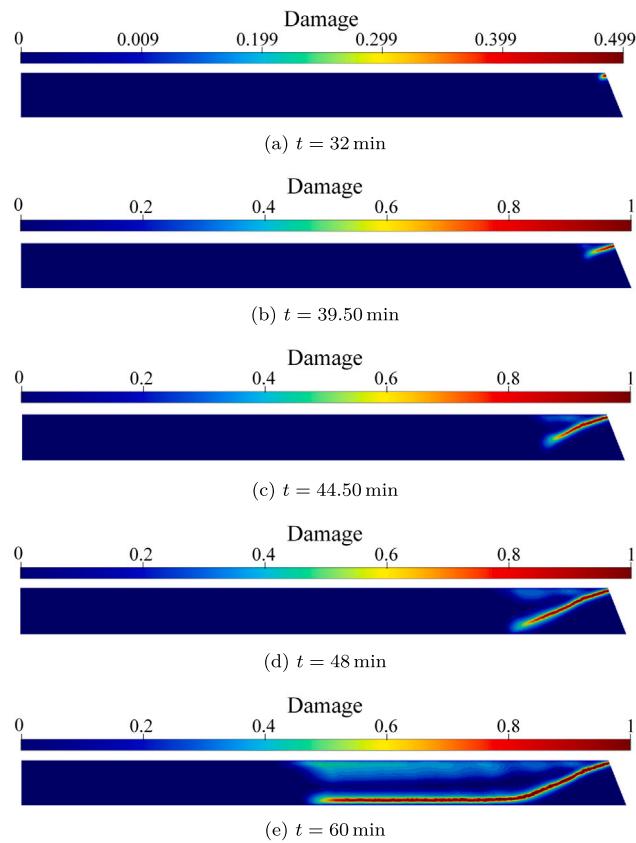


Fig. 34. Crack propagation path in the right portion of the IMC layer with voids.

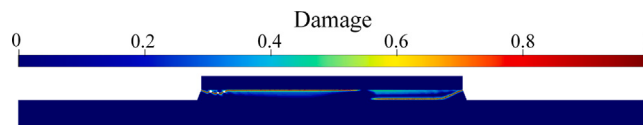


Fig. 35. Final crack pattern of the entire void-containing model, exhibiting a mixed interfacial–bulk fracture mode.

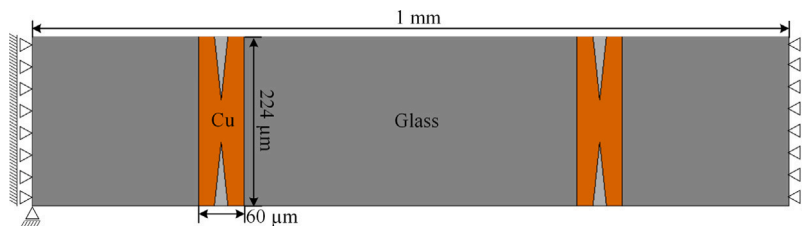


Fig. 36. Geometry and displacement boundary conditions of the considered model; horizontal displacement is constrained on both lateral edges, and vertical displacement is fixed at the lower-left corner.

source of shear stress. In the vertical direction (Fig. 38b), the copper fills exhibit a greater degree of vertical bulging compared to the glass substrate, suggesting a tendency for normal (peel) separation at the interface. The sharp displacement gradients across the copper/glass interface render the TGV opening edges and the copper–glass interfacial regions the most severely deformed zones, thereby providing favorable conditions for subsequent crack initiation and propagation.

To further elucidate the damage mechanisms in the TGVs, Fig. 39 presents the damage contours at four representative time instants. The cracks initiate in the vicinity of the copper/glass interface and subsequently propagate from the outer edges inward, ultimately leading to complete interfacial fracture. This damage evolution corroborates the preceding analysis regarding the role of

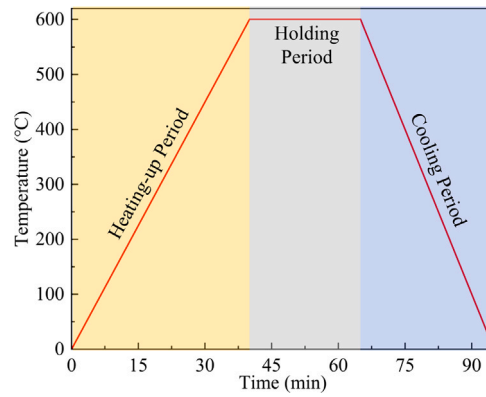


Fig. 37. Prescribed thermal history for the TGV model: heating to 600°C over 40 min, holding until 65 min, and cooling to 0°C by 95 min.

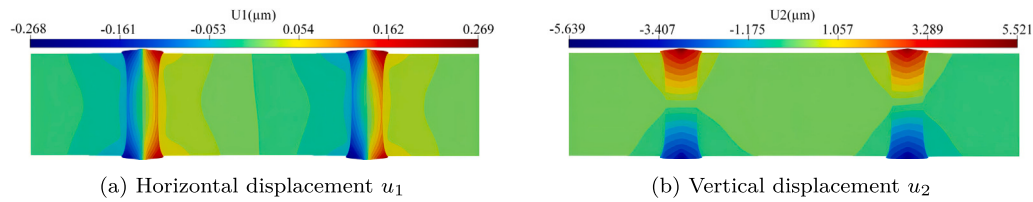


Fig. 38. Displacement fields of the TGV model at $t = 14.25$ min.

displacement incompatibility in driving damage, and the predicted crack patterns are consistent with those reported in the authors' previous PF-CZM-based study [24].

Although the left and right TGVs exhibit the same overall damage-evolution pattern, their local crack paths are not perfectly symmetric. The CTE mismatch between the TGV copper and the glass substrate produces relative sliding and peel deformation at the copper/glass interfaces, as indicated by the displacement fields in Fig. 38. These deformation modes contribute tensile and shear energy to the mixed-mode damage-driving quantity $\bar{Y}$. Once the local driving quantity reaches the damage-initiation threshold $\bar{Y}_0$, stiffness degradation and the resulting stress redistribution promote further damage propagation along locally favorable directions.

The local differences between the left and right crack paths should be interpreted in the context of the discretization and boundary conditions adopted in the present model. The two interfacial regions are discretized using independently generated free meshes, resulting in slight differences in the local mesh arrangement. Moreover, although the horizontal displacement is constrained on both lateral edges, the vertical displacement is fixed only at the lower-left corner to suppress rigid-body motion. The discrete model is therefore not strictly mirror-symmetric. These features may introduce small perturbations in the local damage-driving quantity, which may be amplified through stiffness degradation and stress redistribution during nonlinear damage evolution. Since no dedicated mesh or boundary-condition sensitivity study has been conducted, the observed asymmetry should be interpreted as possible numerical sensitivity rather than as evidence of an intrinsic physical difference between the two sides. Nevertheless, both sides exhibit the same overall interface-dominated damage-evolution mechanism.

5. Conclusions

In this work, the Tpfcm phase-field model has been extended to thermo-mechanically coupled fracture problems in electronic packaging. The proposed framework incorporates thermal strains, a damage-dependent thermal conductivity, and a tension-shear energy decomposition within the Tpfcm formulation, resulting in a three-field coupled model that inherits the length-scale insensitivity of the original Tpfcm, which has been numerically verified through both length-scale parameter and mesh refinement studies. A benchmark comparison between the Tpfcm and the PF-CZM further demonstrates that the Tpfcm reduces both the iteration count and the total computational time, supporting its efficiency advantage in large-scale fracture simulations.

Four representative electronic packaging problems — spanning IMC layer fracture under both mechanical and thermo-mechanical loading, full-IMC solder joint failure, and TGV interfacial fracture — have been investigated. The numerical examples illustrate CTE-mismatch-driven crack propagation in the electronic packaging configurations considered. The adopted tension-shear energy decomposition accounts for shear contributions to the damage-driving force in simulations involving shear-dominated fracture.

The present study has several limitations. All numerical examples employ two-dimensional plane-stress idealizations and therefore do not represent out-of-plane constraint effects, three-dimensional stress redistribution, or spatial crack-front evolution. The model treats material properties as temperature-independent and does not include cycle-dependent fatigue-damage accumulation or separate interface-specific fracture laws. Moreover, the comparison with published experimental observations provides only

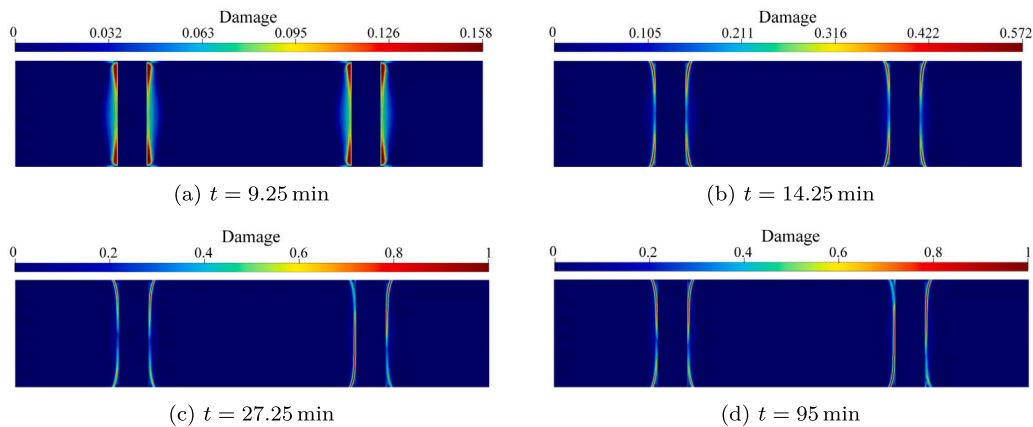


Fig. 39. Damage evolution of the TGV model at four representative time instants.

qualitative morphological corroboration, and systematic quantitative validation under matched conditions remains to be conducted. Accordingly, the results should be interpreted as in-plane fracture trends under the prescribed loading histories rather than as complete three-dimensional or long-term reliability predictions. Future work will focus on three-dimensional modeling, independently calibrated interface fracture properties, temperature-dependent material models, fatigue-damage formulations, and systematic experimental validation.

CRediT authorship contribution statement

Yanpeng Gong: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis. **Letong Zhang:** Writing – review & editing, Visualization, Validation. **Ya Duan:** Supervision. **Tong An:** Supervision. **Xiaoying Zhuang:** Supervision. **Timon Rabczuk:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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